Clarity Pharmaceuticals Ltd

CLINICAL STUDY PROTOCOL: CLP04

Title:	Positron Emission Tomography using ^{64}Cu-SAR-bisPSMA in participants with high-risk prostate cancer prior to radical prostatectomy: A prospective, single-arm, multi-center, blinded-review, Phase 3 diagnostic performance study
Short Title:	CLARIFY
Investigational Product:	^{64}Cu-SAR-bisPSMA
Protocol No.:	CLP04
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IND No.:	158388
Phase:	Phase 3
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Protocol Version (Date):	2.0 (02-Dec-2024)

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SPONSOR PROTOCOL APPROVAL

Protocol Title:	Positron Emission Tomography using ^{64}Cu-SAR-bisPSMA in participants with high-risk prostate cancer prior to radical prostatectomy: A prospective, single-arm, multi-center, blinded-review, Phase 3 diagnostic performance study
Protocol Number:	CLP04
Protocol Version (Date):	2.0 (02-Dec-2024)

This version of the clinical protocol has been approved by Clarity Pharmaceuticals Ltd.

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PROTOCOL SYNOPSIS

Protocol Number	CLP04
Investigational Product	^{64}Cu-SAR-bisPSMA
Protocol Title	Positron Emission Tomography using ^{64}Cu-SAR-bisPSMA in participants with high-risk prostate cancer prior to radical prostatectomy: A prospective, single-arm, multi-center, blinded-review, Phase 3 diagnostic performance study
Short Title	CLARIFY
Phase	Phase 3
IND Number	158388
Study Sponsor	Clarity Pharmaceuticals Ltd.
Study Centers Planned	Multi-center
Study Design	This is a multi-center, single-arm, non-randomized, open-label study of ^{64}Cu-SAR-bisPSMA administered to participants with untreated, histopathology-confirmed prostate cancer (PC), with high-risk features, who are proceeding to radical prostatectomy (RP) with pelvic lymph node dissection (PLND). Participants who provide informed consent will undertake screening visit(s) (Visit 1) to determine their eligibility to participate in the study. The maximum screening period is 28 days and will include institutional Standard of Care (SOC) imaging work up (e.g., magnetic resonance imaging, computed tomography [CT], approved SOC prostate specific membrane antigen [PSMA] positron emission tomography [PET]/CT). Eligible participants will receive a single administration of 200 MBq of ^{64}Cu-SAR-bisPSMA on Day 1 (Visit 2), followed by a PET/CT scan mid-thigh to the skull vertex at 1 to 4 hours post dose (denoted as "Day 1 PET") and at 24 ±6 hours post dose (denoted as "Day 2 PET", Visit 3). Safety assessments will be completed following the administration of ^{64}Cu-SAR-bisPSMA on Day 1, Day 2 and on Day 7 ±4 days (Visit 4) via a telephone call to capture any late-occurring adverse events (AEs). After completion of Visit 4, participants will proceed to RP with PLND. The specimens from the surgery will be processed and analyzed locally. The local pathologist evaluating the samples will be blinded to the ^{64}Cu-SAR-bisPSMA PET/CT scans/results. The histopathology results will be recorded within 90 days of Day 1 (Visit 5 / End of Study Visit). The screening institutional SOC imaging will be interpreted by a single independent, blinded, central reader without access to the ^{64}Cu-SAR-bisPSMA PET/CT scans. The Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT scans will be interpreted locally and by three independent, blinded, central readers. Each reader will assess the scans visually for the presence of abnormal ^{64}Cu-SAR-bisPSMA uptake in the pelvic lymph nodes (LNs), prostate gland, extra-pelvic LNs, visceral/soft tissue and bone. If distant metastatic disease (lesion[s] within the extra-pelvic LNs, bone or visceral/soft tissue regions) is detected on the ^{64}Cu-SAR-bisPSMA PET/CT per the local interpretation of the scans, histopathology (from image-guided biopsy or surgery) should be obtained if clinically feasible.

Primary Objective	**Primary Endpoints**
To assess the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect regional nodal metastases.	Independent primary endpoints for Day 1 and Day 2 PET: • Co-primary endpoint of sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth.

Abbreviations: LN=lymph node; PET=positron emission tomography; PC=prostate cancer

Secondary Objectives	Secondary Endpoints
To investigate the safety and tolerability of ^{64}Cu-SAR-bisPSMA.	Incidence and severity of treatment-emergent AEs and SAEs following administration of ^{64}Cu-SAR-bisPSMA.
To assess the consistency of ^{64}Cu-SAR-bisPSMA PET/CT interpretations for the three central readers.	Inter-reader reliability of agreement estimated with a multiple-reader kappa statistic.
To assess the PPV and NPV of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs.	Reported independently for Day 1 and Day 2 PET: • PPV and NPV of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect PC.	Reported independently for Day 1 and Day 2 PET: • Sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate, pelvic LNs, extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect primary PC.	Reported independently for Day 1 and Day 2 PET: • Sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate gland compared to the Standard of Truth.
To assess the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect regional nodal metastases without subregion matching.	Reported independently for Day 1 and Day 2 PET: • Sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth, without the requirement of subregion (Left or Right) matching between ^{64}Cu-SAR-bisPSMA PET and Standard of Truth.
To determine the dosimetry, PK and excretion of ^{64}Cu-SAR-bisPSMA in participants in the PK substudy.	The following parameters will be expressed from the PET scans: • Absorbed dose (mGy/MBq) in organs. • Effective dose (mSv/MBq). The following parameters will be expressed from blood: • Radioactivity concentration per collection time (MBq/mL), maximum concentration, AUC_{0-t}, AUC_{0-inf}, kel, CL, Vz, T_{max} and $t_{1/2e}$. • Ratios of blood-to-plasma radioactivity at each time point/per participant. • Metabolic profiles (via high-performance liquid chromatography). The following parameters will be expressed from urine: • Radioactivity concentration per collection time (MBq/mL), $Ae_{interval}$, T_{Rmax}, R_{max}, Ae_{0-t}, $\%Fe_{0-t}$. • Metabolic profiles (via high-performance liquid chromatography).

Abbreviations: $\%Fe_{0-t}$=percent of total dose recovered in urine; AE=adverse event; Ae_{0-t}=cumulative amount of excreted radioactivity from time zero to the last collection interval; $Ae_{interval}$=total amount of excreted radioactivity per urine collection interval; AUC_{0-inf}=area under the (activity-time) curve from time of dose administration to infinity; AUC_{0-t}=area under the (activity-time) curve from time of dose to last measurable concentration; CL=total clearance; CT=computed tomography; kel=terminal elimination rate constant; LN=lymph node; NPV=negative predictive value; PET=positron emission tomography; PC=prostate cancer; PK=pharmacokinetic; PPV=positive predictive value; R_{max}=rate max; SAE=serious adverse event; $t_{1/2e}$=elimination half-life; T_{max}=time to reach the maximum concentration; T_{Rmax}=midpoint of the collection interval associated with the maximum observed excretion rate; Vz=total volume of distribution

Exploratory Objectives	**Exploratory Endpoints**
To compare the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect extra-prostatic PC to that of the baseline SOC imaging.	Reported independently for Day 1 and Day 2 PET: • Number and percentage of participants who are SOC imaging pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive. • Number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC imaging pelvic LN negative and histopathology negative. • Number and percentage of participants who are SOC PSMA PET pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive. • Number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC PSMA PET pelvic LN negative and histopathology negative. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC imaging. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC imaging. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC PSMA PET. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC PSMA PET.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect distant metastatic disease (based on local assessment only).	Reported independently for Day 1 and Day 2 PET: • PPV of ^{64}Cu-SAR-bisPSMA PET to detect distant metastatic PC within the extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth.
To assess the number and location of lesion(s) detected on ^{64}Cu-SAR-bisPSMA PET compared to baseline SOC imaging.	Reported independently for Day 1 and Day 2 PET: • Number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC imaging. • Number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC PSMA PET.
To investigate the efficacy of ^{64}Cu-SAR-bisPSMA as a function of baseline variables.	Reported independently for Day 1 and Day 2 PET: • The efficacy parameters of ^{64}Cu-SAR-bisPSMA PET will be assessed as a function of PSA at baseline, testosterone at baseline; Gleason Grade pre-prostatectomy, stage, grade, risk, germline information.
To compare the detection of Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET.	Difference in the total number of lesions detected per participant on Day 1 and Day 2.
To explore the correlation between ^{64}Cu-SAR-bisPSMA PET-positivity and true/false positivity.	Reported independently for Day 1 and Day 2 PET: • Relationship between PET-positivity (SUV, TBR), versus true/false positivity, and as a function of region and size.
To explore the correlation between ^{64}Cu-SAR-bisPSMA PET-positivity and aggressiveness of the disease.	Reported independently for Day 1 and Day 2 PET: • Relationship between PET-positivity (SUV, TBR) and Gleason Grade from the prostatectomy and pelvic LN dissection specimens.
To compare lesion biodistribution and size parameters on Day 1 and	Reported independently for Day 1 and Day 2 PET: • Comparison of SUV and TBR for ^{64}Cu-SAR-bisPSMA PET versus SOC PSMA PET.

Day 2 ^{64}Cu-SAR-bisPSMA PET with SOC PSMA PET.	• Comparison of lesion sizes detected on ^{64}Cu-SAR-bisPSMA PET versus SOC PSMA PET.

Abbreviations: LN=lymph node; PC=prostate cancer; PET=positron emission tomography; PPV=positive predictive value; PSA=prostate-specific antigen; SOC=standard of care; SUV=standardized uptake value; TBR=tumor-to-background ratio

Number of Participants Planned	383
Target Population	The target population for this study is untreated patients with a histopathology-confirmed diagnosis of PC with high-risk features, who are proceeding to RP with PLND.
Inclusion Criteria	1. At least 18 years of age. 2. Signed informed consent. 3. Untreated (i.e. no therapy for PC, including no androgen deprivation therapy of any duration), histologically confirmed adenocarcinoma of the prostate. 4. High-risk or greater PC defined by National Comprehensive Cancer Network Guidelines Version 1.2023 (clinical stage ≥T3a, or Grade Group ≥4, or prostate-specific antigen >20 ng/mL). 5. Patients electing to undergo RP with PLND. 6. For patients who have partners of childbearing potential: partner and/or patient must use a method of birth control with adequate barrier protection.
Exclusion Criteria	1. Administration of any high energy (>300 KeV) gamma-emitting radioisotope within 5 physical half-lives prior to Day 1. 2. Known or expected hypersensitivity to 64Cu-SAR-bisPSMA or any of its components. 3. Patients with known predominant small cell or neuroendocrine PC. 4. History of other active malignancy within 3 years prior to Day 1, except for adequately treated non-melanoma skin cancer and superficial bladder tumors. 5. Any serious medical condition or extenuating circumstance which the investigator feels may interfere with the procedures or evaluations of the study.
Investigational Product, Dose, and Regimen	The investigational product administered in this study is ^{64}Cu-SAR-bisPSMA. ^{64}Cu-SAR-bisPSMA has three basic components: (a) the radionuclide (copper-64) chelated by (b) bisCOSAR (a bifunctional metal chelator) which is linked to (c) two urea based small molecule inhibitors of prostate specific membrane antigen. 200 MBq of ^{64}Cu-SAR-bisPSMA will be administered as a single bolus intravenous injection at Day 1.
Statistical Methods	A Statistical Analysis Plan (SAP) will be written after finalizing the protocol and prior to database lock. The SAP will detail the implementation of all the planned statistical analysis in accordance with the principal features stated in the protocol.

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List of Abbreviations and Definitions of Terms

Abbreviation/Term	Definition
^{18}F	fluorine-18
^{18}F-FDG	^{18}F-fluorodeoxyglucose
^{64}Cu	copper-64
^{64}Cu-SAR-bisPSMA	copper-64 labelled bisCOSAR-(Lys-urea-Glu)$_2$
^{67}Cu	copper-67
^{68}Ga	gallium-68
^{99m}Tc-MDP	^{99m}Tc-methylene diphosphonate
ADL	activities of daily living
ADT	androgen deprivation therapy
AE	adverse events
ATC	Anatomical/Therapeutic/Chemical
BMI	body mass index
CFR	Code of Federal Regulations
CI	confidence interval
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CT	computed tomography
CV%	coefficient of variation as a percentage
DICOM	Digital Imaging and Communications in Medicine
eCRF	electronic case report form
EDC	electronic data capture
eGFR	estimated glomerular filtration rate
FDA	Food and Drug Administration
GCP	Good Clinical Practice
IA	injected activity
IB	Investigator's Brochure
ICF	informed consent form
ICH	International Conference on Harmonisation
IND	Investigational New Drug
ISUP	International Society of Urological Pathology
IV	intravenous
IRB	institutional review board
LN	lymph node
mCRPC	metastatic castrate resistant prostate cancer
MedDRA	Medical Dictionary for Regulatory Activities
MRI	magnetic resonance imaging
^{nat}Cu	natural stable copper
^{nat}Cu-SAR-bisPSMA	^{nat}Cu-labeled SAR-bisPSMA
NCI CTCAE	National Cancer Institute Common Terminology Criteria for Adverse Events
NPV	negative predictive value
PC	prostate cancer
PET	positron emission tomography
PK	pharmacokinetics
PLND	pelvic lymph node dissection
PPV	positive predictive value
PSA	prostate-specific antigen
PSMA	prostate specific membrane antigen

RP	radical prostatectomy
SAE	serious adverse events
SAP	Statistical Analysis Plan
SOC	standard of care
SUV	standardized uptake value
$t_{1/2}$	half-life
$t_{1/2e}$	elimination half-life
TBR	tumor-to-background ratio
TEAE	treatment emergent adverse event
US	United States
WHODrug	World Health Organization drug dictionary

AMENDMENT 1 (V2.0, 02-DEC-2024)

Amendment rationale and summary of changes:

The purpose of this amendment is to incorporate changes described in protocol clarification memos 1 to 5, incorporate new exploratory endpoints, align the protocol with the statistical analysis plan and study manuals, provide some clarifications on study assessments, the patient population, add flexibility for the scheduling of assessments and to address minor typographical errors.

Summary of Changes	
Section Reference Version 2.0, 02-Dec-2024	**Amendment Description**
Throughout the protocol	Version and date of the document has been updated.
Throughout the protocol	Typographical, formatting corrections, administrative changes, and text clarifications.
Sponsor Protocol Approval	Signature page updated to include additional signatories.
Synopsis, Section 5.1.2	Revision of study entry criteria: - Clarification of the meaning of "untreated" prostate cancer included. - Exclusion of participants diagnosed with other malignancies that are expected to alter life expectancy or may interfere with disease assessment. However, participants with a prior history of malignancy that has been adequately treated and who have been disease free for more than 3 years are eligible, as are participants with adequately treated non-melanoma skin cancer, superficial bladder cancer.
Synopsis, Section 3.3, Section 8.3.1.1, Section 11.4.12.4	Additional exploratory endpoints were added to compare the Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET results with the standard of care PSMA PET results. Additional exploratory biodistribution objective and endpoints added to compare the Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET results with SOC PSMA PET.
Synopsis, Section 4.1, Section 9.1.3, Table 5	Day 7 visit window amendment to +/-4 days .
Section 2.2, Section 2.5.2	Addition of ^{18}F-rhPSMA-7.3 (registered as POSLUMA)® to the list of approved PSMA PET agents and inclusion of relevant performance data. Clinical experience with ^{64}Cu-SAR-bisPSMA updated.
Section 4.1, Table 5, Section 8.3.1, Section 9.1.1.1	Inclusion of ^{99m}Tc-MDP bone scan as an example of an institutional standard of care imaging modality.
Section 4.1.1, Table 6, Section 8.7	Day 7 visit (additional assessments for PK substudy) removed.
Section 5.1, Section 8	Clarification added that prospective approval of protocol deviations to recruitment and enrollment criteria, or study assessments, also known as protocol waivers or exemptions, is not permitted.
Section 6.1	Details of ^{64}Cu-SAR-bisPSMA manufacturing standard added.
Section 7.3	Folate supplements and new over the counter medications were removed from the list of prohibited medications.
Table 5, Section 9.1.2.1	Removal of assessment of height from Visit 2.
Table 5	Clarification added that for the calculation of Body Surface Area required for the reporting of eGFR at Visit 3, weight as assessed on Day 1 is to be used.
Table 5, Section 8.3.1	Clarification of the minimum requirements for standard of care imaging at screening provided.
Table 5, Section 8.2.3	Pre-prandial assessment of blood glucose was changed to a random blood glucose assessment.

Summary of Changes	
Section Reference Version 2.0, 02-Dec-2024	**Amendment Description**
Table 5, Section 8.3.1, Section 9.1.1	Clarification added that all participants are required to have baseline SOC imaging. If the required scans were performed within 60 days of Day 1, these scans will serve as baseline SOC imaging and do not need to be repeated at screening. If the required scans are not available, the participant will undergo the missing scan(s) during the Screening period.
Table 5, Section 9.1.2.2, Section 9.1.2.1, Section 9.1.2.2	Modification of assessment windows and schedule of assessments: - Window to perform urinalysis and laboratory assessments on Day 2 (Visit 3) increased to within 9 days after the ^{64}Cu-SAR-bisPSMA administration. This is to allow for decay of radioactivity present in the samples. Due to this change, the Day 7 safety assessments in Table 6 are no longer unique to the Pharmacokinetic substudy and thus were removed from the additional assessments table. - Post-dose urinalysis on Day 1 was removed from the study.
Section 8.3	The requirement to complete all PET scans on the same scanner was clarified.
Section 8.3.2.1	Clarification provided on the definition of a PET-positive lesion.
Section 8.5.1	The Pathology Manual is referenced for specifying data collection requirements from the prostate and pelvic lymph node histology.
Section 8.7.1	Reference to the duplicate assessment of blood, plasma and urine samples was removed duplicates. Correction made to clarify that the metabolic profiles will be assessed by quantitative means.
Section 11.2.5	Definition of the Safety Set was amended to all participants who receive any amount of ^{64}Cu-SAR-bisPSMA regardless of safety assessment status.
Section 11.4.2, Section 11.4.5, Section 11.4.6, Section 11.4.7	Analysis sets used were amended for the summary of demographics and other baseline characteristics, germline information, prostate cancer history, and institutional standard of care imaging.
Section 11.4.12	Uptake of the product will be measured by Standardized Uptake Values (SUV). Specifics to exact SUV parameters reported will be specified in the imaging manuals.
Section 11.4.12.1, Section 11.4.12.2, Section 11.6	Details on the imputation methods for the primary efficacy analyses were removed. These imputation methods will be described in detail in the Statistical Analysis Plan.
Section 11.4.12.2	Details of the sensitivity analysis for the primary efficacy endpoint removed. Full details described in the SAP.
Section 11.4.12.3	Clarification provided on the calculation of sensitivity to detect prostate cancer within the prostate, pelvic LNs, extra-pelvic LNs, bone and visceral/soft tissue regions.
Section 11.6	Clarification added that missing data for adverse event relationship to ^{64}Cu-SAR-bisPSMA will be assumed to be 'related to ^{64}Cu-SAR-bisPSMA' for safety analysis based on the worst-case scenario.
Section 12.4	Clarification added that only United States study sites must forward a signed 1572 Form.
Section 12.7	Clarification on the requirements for the retention of biological samples.
Section 12.7	Mention of physical examination removed.
Section 14	New Investigator Agreement language added for Investigators in Australia.

1 INVESTIGATIONAL TEAM

Refer to the trial specific contact list for details of investigators, sponsor personnel, clinical research organization personnel and facilities used in the study.

2 INTRODUCTION

This study will be conducted in compliance with the protocol, applicable regulatory requirements, and International Conference on Harmonisation (ICH) Good Clinical Practice (GCP) E6 guidelines.

2.1 Background

Prostate cancer (PC) is the second most frequent malignancy occurring in men worldwide[1]. The American Cancer Society estimates ~288,300 new cases of PC and 34,700 deaths from PC in 2023 in the United States (US) alone[2]. Incidence varies and correlates with age, running between 30% for patients 40 to 50 years old to 50 to 80% for patients aged 80 and over[3]. Although the etiology of the disease is unclear, the main risk factors include advanced age, ethnicity, and positive family history[1]. Median age at diagnosis is 67 years and median age at death is 81 years[4,5]. The 5-year relative survival rate for local or regional PC has been over 97% since 2000[6].

While PC-specific mortality has decreased, in recent years there has been a trend toward diagnosis of PC at higher stages, which is presumably due to changes in PC screening practices[7]. The natural history of PC is variable, ranging from indolent tumors that remain confined to the prostate for decades to highly aggressive tumors that rapidly metastasize and lead to death. PC can metastasize throughout the body and is most likely to spread initially to the pelvic lymph nodes (LNs). Once beyond the pelvic LNs, the abdominal LNs and bones are the most common locations of metastasis. More advanced patterns of tumor progression can involve extra-pelvic, non-nodal soft tissue sites like liver and lung. While some patients have indolent disease, where the patient's serum prostate-specific antigen (PSA) values are followed without further treatment, most patients will receive some treatment for their PC. Treatment of PC is largely dependent on the disease stage at initial presentation. At initial presentation roughly 74% of patients have local disease, 13% have regional disease, and 7% have metastatic disease, with the remaining 6% classified as unknown[6]. For patients with grade group 2 or higher PC that is localized to the prostate, local definitive therapy with either radical prostatectomy (RP) or radiation therapy is typically performed.

Prostate specific membrane antigen (PSMA) is a type II transmembrane glycoprotein (glutamate carboxypeptidase type II) that is expressed in normal, benign and malignant prostate tissues, including intraepithelial neoplasia and metastatic specimens[8]. PSMA has two important enzymatic functions; folate hydrolase and N-acetylated-a-linked acidic dipeptidase, and exhibits an endocytic function, in which it spontaneously recycles through endocytic vesicles. Despite its name, PSMA is not specific to the prostate gland and is expressed in other tissues, including salivary and lacrimal glands, duodenal mucosa, proximal renal tubular cells, and subpopulations of neuroendocrine cells in the colonic crypts[9]. PSMA is strongly upregulated in all types of PC, and increases with tumor aggressiveness, metastatic disease, and recurrence, making it an excellent target for both imaging and therapy of PC[10-12].

Clarity Pharmaceuticals Ltd (Sponsor) is developing a radiolabeled inhibitor of PSMA for positron emission tomography (PET) of PSMA-positive lesions in men with PC. The product is copper-64 (^{64}Cu) labelled bisCOSAR-(Lys-urea-Glu)$_2$ (designated as ^{64}Cu-SAR-bisPSMA).

2.2 Imaging of Prostate Cancer

Until recently, imaging of PC has most commonly been carried out by a combination of magnetic resonance imaging (MRI), computed tomography (CT), and ^{99m}Tc-methylene diphosphonate (^{99m}Tc-MDP) bone scan. These traditional imaging modalities, however, suffer

from limited sensitivity and specificity for the detection of disease. This can lead to disease under-staging and the improper selection of treatment. Given the limitations of traditional imaging modalities, the use of molecular imaging using PET has been explored to better identify sites of disease. A number of new PET diagnostic radiopharmaceuticals have therefore emerged in recent years. These include agents that accumulate on the basis of alterations in cellular metabolism as well as those that bind to specific proteins, such as PSMA.

Diagnostic PET radiopharmaceuticals that are commonly used to image PC include ^{11}C-choline, ^{18}F-fluciclovine (registered as AXUMIN®), and more recently, PSMA-based radiopharmaceuticals, ^{68}Ga-PSMA-11 (registered as LOCAMETZ® and ILLUCIX®) and ^{18}F-DCFPyL (registered as PYLARIFY®) and ^{18}F-rhPSMA-7.3 (registered as POSLUMA)®.

Although molecular imaging in oncology is most commonly performed with ^{18}F-fluorodeoxyglucose (^{18}F-FDG) PET, this test has shown limited clinical utility in PC imaging. PC cells typically do not undergo increased aerobic glycolysis, limiting the use of ^{18}F-FDG. These cells do, however, display upregulated *de novo* lipid synthesis and activity of lipogenic enzymes. It is this increase that enables lipid precursors such as choline to function as radiotracers for PET imaging. ^{11}C-choline was approved by the Food and Drug Administration (FDA) for local use by Mayo Clinic in 2012, but choline-based radiotracers are not cancer specific, and can be taken up in other tissues or areas of benign inflammation[13]. In addition, its use is affected by the short half-life ($t_{1/2}$=20 minutes).

In 2016, ^{18}F-fluciclovine was approved by the FDA to evaluate patients with suspected recurrent PC. ^{18}F-fluciclovine targets multiple sodium-dependent and independent channels which are upregulated in PC. ^{18}F-fluciclovine is a synthetic amino acid that is well tolerated and able to detect local and distant PC recurrences across a wide range of PSA values. Similarly, to ^{11}C-choline, uptake of ^{18}F-fluciclovine is not specific to PC and may occur with other types of cancer and benign prostatic hypertrophy. Another limitation of ^{18}F-fluciclovine imaging includes patient preparation, which requires patients not to eat or drink for at least 4 hours before administration of the tracer.

PSMA has been studied extensively as an imaging target in PC. The first commercialized PSMA-targeting imaging agent was ^{111}In-labelled-capromab-pendetide (ProstaScint®), which was approved by the FDA in 1996. However, ProstaScint® was only able to bind to the intracellular epitope of PSMA and was therefore not capable of visualizing viable PC cells leading to poor clinical performance[14]. Since then, the class of PSMA-targeting agents that have been most extensively explored are the urea-based small molecule inhibitors of PSMA. Three of these, ^{68}Ga-PSMA-11 in 2020, ^{18}F-DCFPyL in 2021, and ^{18}F-rhPSMA-7.3 in 2023 have been approved by the FDA for PET imaging of PSMA-positive lesions in men with PC with suspected metastasis who are candidates for initial definitive therapy or with suspected recurrence based on elevated serum PSA level. Evolving evidence demonstrates superior sensitivity and specificity of PSMA PET agents compared to conventional imaging, with frequent identification of sub-centimeter PC lesions. PSMA-targeted PET is effective for imaging disease in the prostate, LNs, soft tissue, and bone in a single examination[15]. In studies comparing ^{11}C-choline and ^{18}F-fluciclovine to ^{68}Ga-PSMA-11, ^{68}Ga-PSMA-11 showed a higher detection rate than ^{11}C-choline PET for LNs as well as bone lesions[16] and ^{18}F-fluciclovine was also shown to be inferior to ^{68}Ga-PSMA-11 in prospective imaging trials[17].

In the Phase 3 trials, ^{68}Ga-PSMA-11, ^{18}F-DCFPyL and ^{18}F-rhPSMA-7.3 demonstrated high specificity (around 90%[18], 96%[19] and 95%[20] respectively) yet a limited sensitivity (around 47%[18], 35%[19] and 27%[20] per approved label, respectively) for the detection of LN metastases in candidates for initial definitive therapy. Thus, while a positive test can be quite helpful in

directing therapy for men with newly diagnosed PC, a negative scan is of limited clinical utility. Therefore, new agents that are able detect and localize small metastatic lesions are highly desirable.

^{64}Cu-SAR-bisPSMA offers a number of advantages over the recently approved PSMA PET agents for PC. These advantages are driven by the longer half-life of ^{64}Cu ($t_{1/2}$=12.7 hours) compared to gallium-68 (^{68}Ga, $t_{1/2}$=68 minutes) and fluorine-18 (^{18}F, $t_{1/2}$=110 minutes) and by the two PSMA-targeting groups (bivalent structure) of SAR-**bis**PSMA – both of which can potentially increase the sensitivity of PSMA PET for the imaging of PC. Unlike the monomeric ^{68}Ga-PSMA-11, ^{18}F-DCFPyL and ^{18}F-rhPSMA-7.3 compounds, ^{64}Cu-SAR-**bis**PSMA has a bivalent structure, which may lead to improved tumor uptake and retention. This was demonstrated in preclinical imaging studies, where ^{64}Cu-SAR-bisPSMA outperformed the monomeric, ^{64}Cu-SAR-PSMA, with respect to both tumor uptake and retention[21]. Due to the short half-life of the isotopes, patients are usually imaged between 50 and 100 minutes after the administration of ^{68}Ga-PSMA-11, ^{18}F-DCFPyL and ^{18}F-rhPSMA-7.3. Imaging later than the recommended imaging window may adversely impact imaging performance due to the decay of the isotope. The longer half-life of ^{64}Cu allows for additional imaging at later time points. The delayed imaging capability allows for flexibility in scheduling of patients, acquisition at multiple timepoints for more accurate dosimetry calculations, and may also enable the detection of additional smaller lesions due to increased uptake driven by the bivalent structure and by an enhanced tumor-to-background ratio (TBR) following biological clearance of the unbound product from the rest of the body over time[22,23]. A Phase 1 trial (CLP03 study, NCT04839367) measured the intensity of tracer uptake (as mean and maximum Standardized Uptake Value [SUV]) and TBR on ^{64}Cu-SAR-bisPSMA PET/CT compared to ^{68}Ga-PSMA-11 PET/CT within the same PC patients. Concordant lesions across the scan pairs demonstrated increased tracer uptake and higher TBR on ^{64}Cu-SAR-bisPSMA compared to ^{68}Ga-PSMA-11 PET/CT, which supports the potential for improved tumor uptake using ^{64}Cu-SAR-bisPSMA. Furthermore, the 12.7 hour half-life of ^{64}Cu also allows production at industrial quantities on cyclotrons, permits centralized manufacturing of ready-to use current Good Manufacturing Practice product to clinical sites and a scalable supply.

2.3 Preclinical Experience

A number of nonclinical studies have been conducted to assess the biodistribution, *in vivo* safety and tolerability as well as tumor imaging and efficacy of SAR-bisPSMA labelled with either natural stable copper (^{nat}Cu), ^{64}Cu or copper-67 (^{67}Cu).

A repeat dose toxicology study of four weekly intravenous (IV) injections of ^{nat}Cu-labeled SAR-bisPSMA (^{nat}Cu-SAR-bisPSMA) in mice resulted in no clinical abnormalities, significant changes in weight or feed intake, and no product related effects in hematology, blood biochemistry and urine analyses. There were no product related macroscopic, nor microscopic findings in organs by histology analysis. The no observed adverse effect level was considered to be 2 mg/kg for ^{nat}Cu-SAR-bisPSMA, which corresponds to a safety factor of the chemical entity, of 192 times the proposed dose of the product[24].

Preclinical imaging studies with ^{64}Cu-SAR-bisPSMA in a PC xenograft model showed high uptake and retention of the product in the tumor over 24 hours (22% injected activity [IA]/g at 1 hour and 26% IA/g at 24 hours post injection)[21]. These results compare favorably to other PSMA-targeted products in similar preclinical models. The preclinical biodistribution of ^{67}Cu-SAR-bisPSMA, the theranostic pair of ^{64}Cu-SAR-bisPSMA, with identical chemical structure but longer half-life ($t_{1/2}$=61.8 hours), showed that the predominant excretion pathway is via renal elimination, with >95% of the ^{67}Cu-SAR-bisPSMA activity in the kidneys clearing within

24 hours. The biodistribution study enabled dosimetry calculations to determine the absorbed radiation dose to organs/tissues for adult males for ^{67}Cu-SAR-bisPSMA and to extrapolate the dosimetry of ^{64}Cu-SAR-bisPSMA. For ^{64}Cu-SAR-bisPSMA, the largest radiation dose is observed in the kidney and urinary bladder, with a value of 0.258 mGy/MBq and 0.059 mGy/MBq, respectively. The effective whole body dose was 0.012 mSv/MBq, which for an administration of 200 MBq of ^{64}Cu-SAR-bisPSMA would result in an effective whole body dose of 2.43 mSv[25].

In summary, the nonclinical data indicate that ^{64}Cu-SAR-bisPSMA binds to PSMA-expressing tumors with high affinity, appears to be well-tolerated and has the potential to be used to obtain high quality images of PSMA-expressing tumors in human cancers.

Please refer to the current ^{64}Cu-SAR-bisPSMA Investigator's Brochure (IB) for complete details of completed nonclinical studies.

2.4 Clinical Experience

A Phase 1 study of ^{64}Cu-SAR-bisPSMA (CLP03) has been completed, a Phase 1/2a study of ^{64}Cu-SAR-bisPSMA (CLP06) has been completed (clinical study report pending) and one study utilizing ^{64}Cu-SAR-bisPSMA is in progress (CLP05).

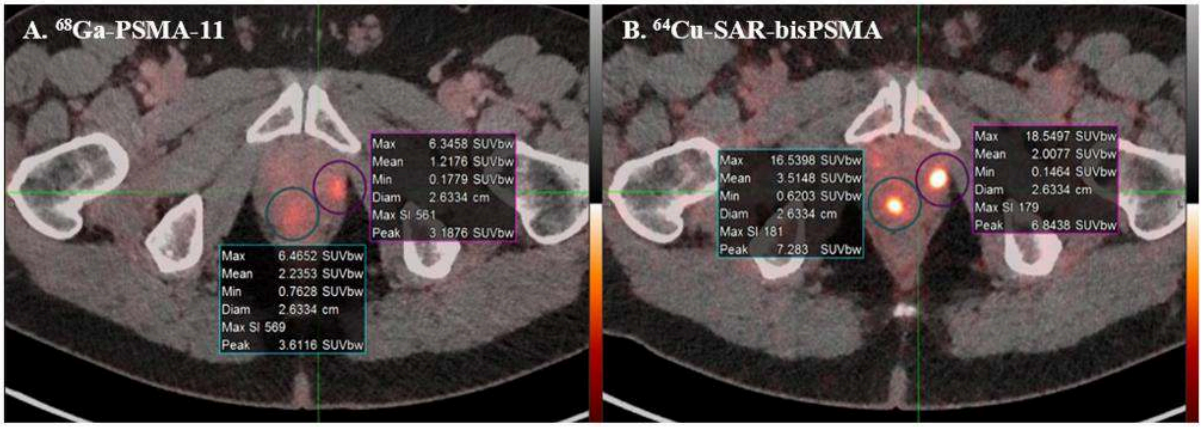
The CLP03 study (NCT04839367) was completed under Australia's Therapeutic Goods Administration Clinical Trial Notification scheme at three investigative sites in Australia. This was a prospective, Phase 1, multi-center, blinded review, dose-ranging, non-randomized study of ^{64}Cu-SAR-bisPSMA in 30 participants with untreated, confirmed PC, with intermediate- to high-risk features prior to RP with pelvic lymph node dissection (PLND). The aims of the study were to evaluate safety and tolerability of ^{64}Cu-SAR-bisPSMA, assess efficacy to detect primary PC and evaluate image quality at various dose levels. Screening standard of care (SOC) imaging included a ^{68}Ga-PSMA-11 PET/CT (at 45 to 60 minutes post dose per institutional protocols) acquired within 5 weeks, but not closer than 6 hours prior to the administration of ^{64}Cu-SAR-bisPSMA. Enrolled participants were sequentially allocated to one of three dose cohorts to receive either 100 MBq (n=6), 150 MBq (n=6) or 200 MBq (n=18) of ^{64}Cu-SAR-bisPSMA on Day 0, followed by a whole body PET/CT at 3 ±1 hour post dose. Safety was assessed pre and post dose on Day 0, Day 1, between Day 5 to Day 28 and following the completion of prostatectomy up to 11 weeks post Day 0. Participants proceeded to RP following the completion of the Day 5 to Day 28 safety visit. The ^{64}Cu-SAR-bisPSMA and pre-study ^{68}Ga-PSMA-11 PET/CT scans were interpreted in a randomized order by two independent, blinded, central readers. The intensity of tracer uptake was compared on the ^{64}Cu-SAR-bisPSMA and ^{68}Ga-PSMA-11 PET/CT in up to 5 concordant lesions per participant.

A single dose of ^{64}Cu-SAR-bisPSMA was found to be safe and well tolerated when administered to participants with confirmed PC by IV injection. There were no ^{64}Cu-SAR-bisPSMA related serious adverse events (SAEs) or deaths, or adverse events (AEs) leading to premature discontinuation. There was only one AE that was considered related to ^{64}Cu-SAR-bisPSMA by the Investigator, which was Grade 1 dysgeusia (occasional metallic taste) in a participant enrolled in the 200 MBq ^{64}Cu-SAR-bisPSMA cohort. For the primary efficacy endpoint, in the 200 MBq cohort, ^{64}Cu-SAR-bisPSMA had a high true positive rate for the detection of primary PC confirmed by histopathology (100% by Reader 1 and 85.7% by Reader 2), and low false negative rate (0% by Reader 1 and 14.3% by Reader 2). Both readers independently scored 200 MBq of ^{64}Cu-SAR-bisPSMA as the dose level providing the highest image quality. This is the dose level that will be utilized in this study (refer to Section 2.5.2). In the 200 MBq cohort, there were more PET-positive lesions detected by ^{64}Cu-SAR-bisPSMA

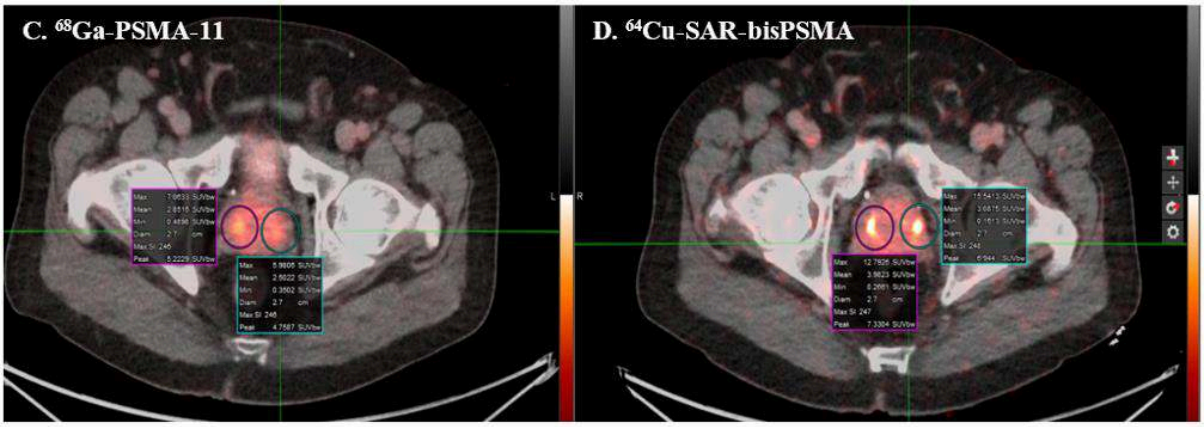
compared to ^{68}Ga-PSMA-11 PET/CT by both central readers (Reader 1 identified 41 lesions using ^{64}Cu-SAR-bisPSMA and 36 lesions using ^{68}Ga-PSMA-11, while Reader 2 identified 31 lesions using ^{64}Cu-SAR-bisPSMA and 22 lesions using ^{68}Ga-PSMA-11). Concordant lesions identified on ^{64}Cu-SAR-bisPSMA and ^{68}Ga-PSMA-11 PET/CT showed consistently higher uptake and TBR on ^{64}Cu-SAR-bisPSMA compared to ^{68}Ga-PSMA-11. Intra-individual PET/CT images from two participants in the CLP03 study in Figure 1 demonstrate higher uptake and clearer delineation of lesions on 200 MBq of ^{64}Cu-SAR-bisPSMA (B, D) compared to ^{68}Ga-PSMA-11 (A, C). Note, the intra-individual image pairs were acquired on the same scanner.

Figure 1: Intra-individual PET/CT images from two participants comparing 200 MBq of ^{64}Cu-SAR-bisPSMA (B, D) and ^{68}Ga-PSMA-11 (A, C)

Participant #1



Participant #2



The CLP05 clinical study (NCT04868604) a is prospective, Phase 1/2a, multi-center, open-label, non-randomized, dose-escalation, dose-finding, cohort expansion, theranostic study of ^{64}Cu-SAR-bisPSMA and ^{67}Cu-SAR-bisPSMA administered to participants with metastatic castrate resistant prostate cancer (mCRPC) being conducted at a number of clinical sites in the US. ^{64}Cu-SAR-bisPSMA PET/CT is used for identification and ^{67}Cu-SAR-bisPSMA for the treatment of PSMA-expressing lesions. The study consists of a Dosimetry Phase, a Dose Escalation Phase and a Cohort Expansion Phase. The Dosimetry Phase of the study is now completed. This phase investigated the safety and radiation dosimetry of 200 MBq of ^{64}Cu-SAR-bisPSMA in 6 participants. A single dose of 200 MBq of ^{64}Cu-SAR-bisPSMA was well tolerated and resulted in a relatively low effective dose (refer to Section 2.5.2).

The CLP06 clinical study (NCT05249127) is a prospective, Phase 1/2, multi-center, blinded review, non-randomized study of ^{64}Cu-SAR-bisPSMA PET/CT in 50 participants with biochemical recurrence of PC following initial definitive therapy. The primary objective of the study is to assess the safety and preliminary efficacy of ^{64}Cu-SAR-bisPSMA PET/CT to localize the site(s) of PC recurrence across two PET acquisition timepoints (at 1 to 4 hours and 24 hours post dose).

The clinical data obtained indicate that PET with ^{64}Cu-SAR-bisPSMA is well tolerated, with no reported SAE deemed related to ^{64}Cu-SAR-bisPSMA in any of the studies described above at any dose level tested. Furthermore, in the completed CLP03 Phase 1 clinical trial, ^{64}Cu-SAR-bisPSMA has demonstrated a high true positive rate for the detection of primary PC, showed a greater number of PET-positive lesions, higher tracer accumulation and TBR compared to ^{68}Ga-PSMA-11 PET/CT in concordant lesions. Thus, ^{64}Cu-SAR-bisPSMA PET/CT promises to be a useful diagnostic agent to image PC in patients with suspected metastasis who are candidates for initial definitive therapy.

Please refer to the current ^{64}Cu-SAR-bisPSMA IB for details on the completed clinical studies.

2.5 Rationale for the Study and Study Design

Several imaging modalities are currently employed for the diagnosis, staging and prognosis of PC metastases, however, none have optimal performance, particularly when it comes to the detection of small pelvic LN metastases in patients newly diagnosed with PC. The need for improved PC imaging is reflected in the recurrence rate of 20 to 40% following definitive therapy (with RP or radiation therapy) of what was thought to be disease localized to the prostate gland[26]. The nonclinical and clinical data obtained to date indicate that PET with ^{64}Cu-SAR-bisPSMA is well tolerated, with high true positive rate for the detection of primary PC. The number of PET-positive lesions identified, tracer uptake and TBR of ^{64}Cu-SAR-bisPSMA PET/CT all compared favorably to that of ^{68}Ga-PSMA-11 PET/CT when studied in patients with untreated, histopathology-confirmed PC, proceeding to RP with PLND (refer to Section 2.4). These findings warrant further investigation in a Phase 3 clinical study.

2.5.1 Patient Population/Endpoints

This study is a prospective, non-randomized, single-arm, open-label, multi-reader, multi-center, Phase 3 diagnostic performance study of ^{64}Cu-SAR-bisPSMA PET/CT. The aim of the study is to evaluate the safety and determine the sensitivity and specificity of ^{64}Cu-SAR-bisPSMA to detect PC within the pelvic LNs at two PET acquisition timepoints, independently. The target study population is patients with high risk, untreated PC, who are proceeding to RP with PLND. Investigating ^{64}Cu-SAR-bisPSMA PET/CT in this population is reasonable as PSMA PET already plays an important role in the diagnostic workup of newly diagnosed patients who are candidates for initial definitive therapy. As PC most often spreads to the pelvic LNs before becoming widely metastatic, assessing performance of ^{64}Cu-SAR-bisPSMA PET/CT for detection of pelvic LN involvement as an endpoint is both clinically relevant and practical. Focusing on a subset of patients undergoing surgery (as opposed to all patients being staged prior to definitive therapy) is deemed reasonable due to the need to obtain a histological confirmation of LN involvement that enables the assessment of sensitivity and specificity. The additional diagnostic efficacy assessments used in this study (e.g., positive and negative predictive values) are widely used and considered appropriate for the evaluation of performance of diagnostic imaging agents. The collection of standard safety parameters used in this study such as AE reporting, assessment of vital signs, clinical laboratory, and electrocardiograms (ECGs) are considered appropriate for monitoring the health and well-

being of participants in late phase clinical studies. The biodistribution, elimination and dosimetry assessments were selected to further characterize the pharmacokinetics (PK) of ^{64}Cu-SAR-bisPSMA in humans.

This study will provide the Sponsor with information on the safety, efficacy and PK of ^{64}Cu-SAR-bisPSMA to further clinical development and support registration of ^{64}Cu-SAR-bisPSMA PET/CT as a radioactive diagnostic agent for PET of PSMA-positive lesions in men with PC.

2.5.2 Dose of ^{64}Cu-SAR-bisPSMA

^{64}Cu-SAR-bisPSMA will be administered at a fixed administered activity of 200 MBq.

This activity yielded the highest image quality score in the dose ranging CLP03 Phase 1 study in participants with known primary PC according to both independent, blinded, central readers for PET at 2 to 4 hours post dose. In terms of efficacy, the rate of participants deemed positive for primary PC for the 100, 150 and 200 MBq cohort was 100%, 83%, 100% according to Reader 1 and 33%, 17% and 67% according to Reader 2, respectively.

This activity is also utilized in the ongoing clinical studies with ^{64}Cu-SAR-bisPSMA (CLP05 and CLP06) and preliminary data from these investigations also support the use of 200 MBq. The CLP06 Phase 1/2 study investigates the same PET acquisition timepoints as this study (i.e., 1 to 4 hours and at 24 ±6 hours post dose). Images obtained across multiple investigative sites demonstrate that 200 MBq of ^{64}Cu-SAR-bisPSMA is an adequate activity to allow sufficient image quality not only for the 1 to 4 hour timepoint but also for delayed imaging at 24 hours post dose using standard PET acquisition protocols.

As part of the CLP05 Phase 1/2a trial, human biodistribution studies were used to estimate the organ radiation dosimetry of ^{64}Cu-SAR-bisPSMA in adult males. An administration of 200 MBq of ^{64}Cu-SAR-bisPSMA enabled imaging out to 72 hours and resulted in a relatively low effective dose. According to the radiation dosimetry estimates obtained, the whole body effective dose to an adult male weighing 73 kg is 5.7 mSv for a 200 MBq injection of ^{64}Cu-SAR-bisPSMA. A whole body effective dose of this magnitude is comparable to other routinely used imaging agents such as ^{18}F-FDG (8.4 mSv according to the Nuclear Medicine Radiation Dose Tool - Society of Nuclear Medicine and Molecular Imaging), and the recently approved ^{68}Ga-PSMA-11, ^{18}F-DCFPyL and ^{18}F-rhPSMA-7.3 at the higher end of the recommended administered activity (4.4 mSv[18], 4.3 mSv[19] and 4.1 mSv[20], respectively).

The safety data from the completed and ongoing studies indicate that this dose level is well tolerated, with no reported SAEs deemed related to the investigational product in any of the studies.

Based on the overall safety, efficacy, dosimetry, and image quality results obtained to date, 200 MBq of ^{64}Cu-SAR-bisPSMA is deemed the optimal dose level for this protocol for acquisition of PET at 1 to 4 hours and at 24 ±6 hours post dose.

2.5.3 Imaging Timepoint Selection

A PET/CT scan will be acquired between 1 to 4 hours post administration of ^{64}Cu-SAR-bisPSMA (denoted as “Day 1 PET”). This window is considered practical, provides flexibility, and allows clinicians to follow clinical routine already in place for PSMA PET agents (for ^{68}Ga-PSMA-11 and ^{18}F-DCFPyL, PET acquisition is initiated approximately 1 hour post administration). A second PET/CT at 24 ±6 hours post dose (denoted as “Day 2 PET”) will also be completed as delayed imaging may enable detection of additional smaller lesions[22] due

to enhanced TBRs following biological clearance of the unbound product from the rest of the body over time and continued tumor uptake of SAR-bisPSMA.

Small animal PET/CT images of LNCaP tumor-bearing NSG mice at 1, 4 and 24 hours post administration of ^{64}Cu-SAR-bisPSMA showed that the tumor is clearly visible at 1 hour post dose with an average TBR of 35.3 ±7.3. TBR at 4 and 24 hours was much higher than at 1 hour (72.1 ±14.1 and 104.6 ±17.7, respectively), which demonstrates a high degree of accumulation and retention in tumor and clearance from background over time27.

2.6 Benefit/Risk Assessment

Benefit: It is not known if study participants will gain any improvement in their own healthcare outcomes as a result of their participation in this study. A general societal benefit is widely considered to accrue from medical scientific knowledge accumulated through the conduct of ethical research on participants.

It is hypothesized that due to the longer half-life of ^{64}Cu and the dual targeting of PSMA, ^{64}Cu-SAR-bisPSMA will enable acquisition of PET scans at delayed imaging timepoints. Delayed imaging has the potential to improve lesion detection and allows clinics flexibility in scheduling of the scans. It also improves patient's access to care in areas with limited access to the current short half-life PSMA PET tracers, including but not limited to poorly serviced areas and underserved, underrepresented populations.

Risk: All study participants will receive a single administration of ^{64}Cu-SAR-bisPSMA. Specific foreseeable risks to study participants include:

- Radiation dose received by the study participant: Human biodistribution studies in patients with mCRPC estimate that an administration of 200 MBq of ^{64}Cu-SAR-bisPSMA would result in an effective dose of 5.7 mSv (i.e., 0.0285 mSv/MBq), which is comparable to the effective doses of other routinely used imaging agents at their recommended administered activity. There is a potential hazard from the additional radiation burden to study participants. The study also involves low dose CT scan(s) performed for attenuation correction of the PET scan(s) and for the anatomical correlation to allow more precise delineation of organs.

- Chemical toxicity of ^{64}Cu-SAR-bisPSMA and other constituents of the injection: The constituents (including excipients and stabilizers) of ^{64}Cu-SAR-bisPSMA are at concentrations that are known to be safe for human use. The investigational product, as supplied, will result in a maximum quantity not exceeding 50 µg of peptide mass per participant for an administration. The administered peptide mass is therefore at least 192 times less than the dose of ^{nat}Cu-SAR-bisPSMA that has been demonstrated to be without any relevant toxicity in rodent toxicology studies (based on a 60 kg subject)[24]. The administered dose is well below the limits specified by the FDA for microdosing studies.

- Intravenous administration: Minor pain may occur at the time of venipuncture and minor bruising may also result; however, patients are exposed to these same risks during routine care. Whilst extravasation of PET radiopharmaceuticals is possible, it is unlikely to be associated with adverse outcomes due to the very small mass of materials contained in the injection, and the small amount of radioactivity available for injection. Nevertheless, an indwelling canula can be established prior to investigational product administration to mitigate this risk.

- Incidental findings of possible clinical significance: Appropriate diagnostic testing can assist in making a correct diagnosis and in providing additional information to guide patient management. An incorrect diagnosis (e.g., false negative or false positive results of the diagnostic tests) can have important consequences in patient management. An incidental finding is an imaging finding of potential health importance concerning the study participant, which is discovered while conducting imaging research, but is beyond the aims of the study. It is the responsibility of the local site or of the study Investigator to arrange for routine clinical review of study scans and clinical follow-up, as required. Policies for the communication of the incidental finding to the research participant should align with national regulations and customs. As part of the informed consent process, research participants will be informed about the possibility that incidental imaging findings may be detected, and about the pathway for handling such findings.

3 STUDY OBJECTIVES AND ENDPOINTS

3.1 Primary Objective and Endpoints

The primary study objective and endpoints to be evaluated in this study are presented in Table 1.

Table 1 Primary Study Objective and Endpoints

Objective	Endpoint
To assess the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect regional nodal metastases.	Independent primary endpoints for Day 1 and Day 2 PET: • Co-primary endpoint of sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth.

Abbreviations: LN=lymph node; PET=positron emission tomography; PC=prostate cancer

3.2 Secondary Objectives and Endpoints

The secondary study objectives and endpoints to be evaluated in this study are presented in Table 2.

Table 2 Secondary Study Objectives and Endpoints

Objective	Endpoint
To investigate the safety and tolerability of ^{64}Cu-SAR-bisPSMA.	Incidence and severity of treatment-emergent AEs and SAEs following administration of ^{64}Cu-SAR-bisPSMA.
To assess the consistency of ^{64}Cu-SAR-bisPSMA PET/CT interpretations for the three central readers.	Inter-reader reliability of agreement estimated with a multiple-reader kappa statistic.
To assess the PPV and NPV of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs.	Reported independently for Day 1 and Day 2 PET: • PPV and NPV of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect PC.	Reported independently for Day 1 and Day 2 PET: • Sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate, pelvic LNs, extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect primary PC.	Reported independently for Day 1 and Day 2 PET: • Sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate gland compared to the Standard of Truth.
To assess the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect regional nodal metastases without subregion matching.	Reported independently for Day 1 and Day 2 PET: • Sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth, without the requirement of subregion (Left or Right) matching between ^{64}Cu-SAR-bisPSMA PET and Standard of Truth.
To determine the dosimetry, PK and excretion of ^{64}Cu-SAR-bisPSMA in participants in the PK substudy.	The following parameters will be expressed from the PET scans: • Absorbed dose (mGy/MBq) in organs. • Effective dose (mSv/MBq). The following parameters will be expressed from blood:

	• Radioactivity concentration per collection time (MBq/mL), maximum concentration, AUC_{0-t}, AUC_{0-inf}, kel, CL, Vz, T_{max} and $t_{1/2e}$. • Ratios of blood-to-plasma radioactivity at each time point/per participant. • Metabolic profiles (via high-performance liquid chromatography). The following parameters will be expressed from urine: • Radioactivity concentration per collection time (MBq/mL), $Ae_{interval}$, T_{Rmax}, R_{max}, Ae_{0-t}%Fe_{0-t}. • Metabolic profiles (via high-performance liquid chromatography).

Abbreviations: %Fe_{0-t}=percent of total dose recovered in urine; AE=adverse event; Ae_{0-t}=cumulative amount of excreted radioactivity from time zero to the last collection interval; $Ae_{interval}$=total amount of excreted radioactivity per urine collection interval; AUC_{0-inf}=area under the (activity-time) curve from time of dose administration to infinity; AUC_{0-t}=area under the (activity-time) curve from time of dose to last measurable concentration; CL=total clearance; CT=computed tomography; kel=terminal elimination rate constant; LN=lymph node; NPV=negative predictive value; PET=positron emission tomography; PC=prostate cancer; PK=pharmacokinetic; PPV=positive predictive value; R_{max}=rate max; SAE=serious adverse event; $t_{1/2e}$=elimination half-life; T_{max}=time to reach the maximum concentration; T_{Rmax}=midpoint of the collection interval associated with the maximum observed excretion rate; Vz=total volume of distribution

3.3 Exploratory Objectives and Endpoints

The exploratory study objectives and endpoints to be evaluated in this study are presented in Table 3.

Table 3 Exploratory Study Objectives and Endpoints

Objective	Endpoint
To compare the diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect extra-prostatic PC to that of the baseline SOC imaging.	Reported independently for Day 1 and Day 2 PET: • Number and percentage of participants who are SOC imaging pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive. • Number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC imaging pelvic LN negative and histopathology negative. • Number and percentage of participants who are SOC PSMA PET pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive. • Number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC PSMA PET pelvic LN negative and histopathology negative. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC imaging. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC imaging. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC PSMA PET. • Percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC PSMA PET.
To assess the ability of ^{64}Cu-SAR-bisPSMA PET to detect distant metastatic disease (based on local assessment only).	Reported independently for Day 1 and Day 2 PET: • PPV of ^{64}Cu-SAR-bisPSMA PET to detect distant metastatic PC within the extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth.

To assess the number and location of lesion(s) detected on ^{64}Cu-SAR-bisPSMA PET compared to baseline SOC imaging.	Reported independently for Day 1 and Day 2 PET: • Number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC imaging. • Number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC PSMA PET.
To investigate the efficacy of ^{64}Cu-SAR-bisPSMA as a function of baseline variables.	Reported independently for Day 1 and Day 2 PET: • The efficacy parameters of ^{64}Cu-SAR-bisPSMA PET will be assessed as a function of PSA at baseline, testosterone at baseline; Gleason Grade pre-prostatectomy, stage, grade, risk, germline information.
To compare the detection of Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET.	Difference in the total number of lesions detected per participant on Day 1 and Day 2.
To explore the correlation between ^{64}Cu-SAR-bisPSMA PET-positivity and true/false positivity.	Reported independently for Day 1 and Day 2 PET: • Relationship between PET-positivity (SUV, TBR), versus true/false positivity, and as a function of region and size.
To explore the correlation between ^{64}Cu-SAR-bisPSMA PET-positivity and aggressiveness of the disease.	Reported independently for Day 1 and Day 2 PET: • Relationship between PET-positivity (SUV, TBR) and Gleason Grade from the prostatectomy and pelvic LN dissection specimens.
To compare lesion biodistribution and size parameters on Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET with SOC PSMA PET.	Reported independently for Day 1 and Day 2 PET: • Comparison of SUV and TBR for ^{64}Cu-SAR-bisPSMA PET versus SOC PSMA PET. • Comparison of lesion sizes detected on ^{64}Cu-SAR-bisPSMA PET versus SOC PSMA PET.

Abbreviations: LN=lymph node; PC=prostate cancer; PET=positron emission tomography; PPV=positive predictive value; PSA=prostate-specific antigen; SOC=standard of care; SUV=standardized uptake value; TBR=tumor-to-background ratio

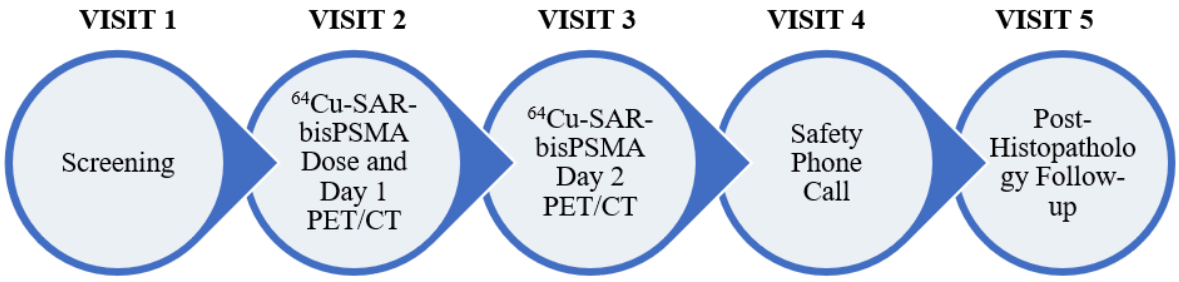
4 STUDY DESIGN

4.1 Study Overview

This is a prospective, non-randomized, single-arm, open-label, multi-reader, multi-center, Phase 3 study designed to evaluate the safety and diagnostic performance of ^{64}Cu-SAR-bisPSMA PET/CT in participants with untreated, histopathology-confirmed PC, with high-risk features, who are proceeding to RP with PLND. A total of 383 participants will be enrolled in this study.

A schematic diagram of the study design is presented in Figure 2. Details of study procedures during each visit are described in Table 5.

Figure 2 Overall Study Design



Abbreviations: CT=computed tomography; PET=positron emission tomography

Participants who provide informed consent will undertake screening visit(s) (Visit 1) to determine their eligibility to participate in the study. The maximum screening period is 28 days and will include institutional SOC imaging work up (e.g., MRI, CT, approved SOC PSMA PET/CT, with or without ^{99m}Tc-MDP bone scan). Eligible participants will receive a single administration of 200 MBq of ^{64}Cu-SAR-bisPSMA on Day 1 (Visit 2), followed by a PET/CT scan mid-thigh to the skull vertex at 1 to 4 hours post dose (denoted as "Day 1 PET") and at 24 ±6 hours post dose (denoted as "Day 2 PET", Visit 3). Safety assessments will be completed following the administration of ^{64}Cu-SAR-bisPSMA on Day 1, Day 2 and on Day 7 ±4 days (Visit 4) via a telephone call to capture any late-occurring AEs. After completion of Visit 4, participants will proceed to RP with PLND. The specimens from the surgery will be processed and analyzed locally. The local pathologist evaluating the samples will be blinded to the ^{64}Cu-SAR-bisPSMA PET/CT scans/results. For participants who do not proceed to RP with PLND, obtaining histopathology from excisional biopsy or laparoscopic PLND will be attempted, if this is clinically feasible prior to the participant undergoing any alternate intervention for PC to enable the assessment of the primary endpoint. For these participants the alternate intervention for PC will also be recorded. The histopathology results will be recorded within 90 days of Day 1 (Visit 5 / End of Study Visit).

The screening institutional SOC imaging will be interpreted by a single independent, blinded, central reader without access to the ^{64}Cu-SAR-bisPSMA PET/CT scans. The Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT scans will be interpreted locally and by three independent, blinded, central readers. Each reader will assess the scans visually for the presence of abnormal ^{64}Cu-SAR-bisPSMA uptake in the pelvic LNs, prostate gland, extra-pelvic LNs, visceral/soft tissue and bone. If distant metastatic disease (lesion[s] within the extra-pelvic LNs, bone or visceral/soft tissue regions) is detected on the ^{64}Cu-SAR-bisPSMA PET/CT per the **local**

interpretation of the scans, histopathology (from image-guided biopsy or surgery) should be obtained if clinically feasible.

4.1.1 Pharmacokinetic Substudy

This study will also include a PK substudy to provide a more complete assessment of biodistribution, elimination and dosimetry of ^{64}Cu-SAR-bisPSMA in humans. Approximately 1 to 2 clinical sites will be selected to participate in the PK substudy. A subset of 10 participants enrolled at these sites will be included in the PK substudy. These participants will complete additional assessments on Day 1, Day 2, and Day 3 as per Table 6.

4.2 Study Duration, Beginning and End of Study

Participants are considered enrolled in the study once they are appropriately consented and screened, with eligibility verified by an Investigator.

The expected study recruitment period is ~12 months.

The total trial duration is ~16 months.

The expected maximum study involvement for a participant is ~ 4 months including:

- Screening period of up to 28 days;
- On-study period of up to 90 days.

The end of study is defined as the date of the final participant visit/contact. In addition, the Sponsor may decide to terminate the study at any time (see Section 5.5).

5 STUDY ENROLLMENT AND WITHDRAWAL

5.1 Study Population

The target population for this study is untreated patients with a histopathology-confirmed diagnosis of PC with high-risk features, who are proceeding to RP with PLND. The inclusion and exclusion criteria are outlined below.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1.1 Inclusion Criteria

Study candidates must meet all the following criteria to be eligible for the study:

1. At least 18 years of age.
2. Signed informed consent.
3. Untreated (i.e. no therapy for PC, including no androgen deprivation therapy [ADT] of any duration), histologically confirmed adenocarcinoma of the prostate.
4. High-risk or greater PC defined by National Comprehensive Cancer Network Guidelines Version 1.2023[28] (clinical stage ≥T3a, or Grade Group ≥4, or PSA >20 ng/mL).
5. Patients electing to undergo RP with PLND.
6. For patients who have partners of childbearing potential: partner and/or patient must use a method of birth control with adequate barrier protection.

5.1.2 Exclusion Criteria

Study candidates who meet any of the following criteria are not to be enrolled in the study:

1. Administration of any high energy (>300 KeV) gamma-emitting radioisotope within 5 physical half-lives prior to Day 1.
2. Known or expected hypersensitivity to ^{64}Cu-SAR-bisPSMA or any of its components.
3. Patients with known predominant small cell or neuroendocrine PC.
4. History of other active malignancy within 3 years prior to Day 1, except for adequately treated non-melanoma skin cancer and superficial bladder tumors.
5. Any serious medical condition or extenuating circumstance which the Investigator feels may interfere with the procedures or evaluations of the study.

5.2 Enrollment and Method of Assigning Participants to Treatment Groups

This is a non-randomized, single-arm study. After completing the screening period, participants who meet all inclusion criteria and no exclusion criteria, will be enrolled into the study and receive a single administration of ^{64}Cu-SAR-bisPSMA.

A total of 383 participants will be enrolled into the study.

Each participant will be identified in the study by a Participant Number, that is assigned when the participant is first entered into Screening and is retained as the primary identifier for the participant throughout his entire participation in the study. The Participant Number consists of

the Centre Number (as assigned by the Sponsor to the investigative site) with a sequential Participant Number suffixed to it, so that each participant is numbered uniquely across the entire database. Upon signing the informed consent form (ICF), the participant is assigned to the next sequential Participant Number available to the Investigator.

5.3 Study Blinding

In this study all participants will receive ^{64}Cu-SAR-bisPSMA and thus blinding of investigational product is not applicable.

To allow for a non-biased assessment of the ^{64}Cu-SAR-bisPSMA PET/CT scans, the three independent, central readers will be blinded to the participants-specific information (identifiers, medical history, laboratory results), the timepoint of the scan (i.e., Day 1 or Day 2), results of evaluations as part of the Standard of Truth, results of other imaging modalities, final diagnosis / outcome, and details of the protocol.

To allow for the non-biased assessment of the histopathology Standard of Truth, the Investigator must ensure that the local pathologist evaluating the samples for the primary endpoint is blinded to the ^{64}Cu-SAR-bisPSMA PET/CT scans/results.

The screening SOC imaging will be interpreted by a single, independent, central reader (distinct from the three readers evaluating the ^{64}Cu-SAR-bisPSMA PET/CT scans) who will be blinded to the ^{64}Cu-SAR-bisPSMA PET/CT scans/results, participants-specific information (identifiers, medical history, laboratory results), results of evaluations as part of the Standard of Truth, results of other imaging modalities, final diagnosis / outcome, and details of the protocol. This is to allow for a non-biased comparison of the number/location of lesions identified on the SOC imaging and ^{64}Cu-SAR-bisPSMA PET/CT.

5.4 Participant Withdrawal and Replacement

5.4.1 Screening Failures and Rescreening

5.4.1.1 Screen Failures

Screen failures are those participants who do not meet the study eligibility criteria. For participants who are considered a screen failure, the reason for screen failure will be entered on the Screening Log. The demographic information, informed consent, inclusion/exclusion and End of Study Visit pages must also be completed for screen failure participants. If the participant experiences an SAE during the screening period, this information must also be entered (see Section 10.3 for SAE reporting details).

5.4.1.2 Rescreening

Participants who fail screening assessments may be re-screened. Rescreening and the specific assessments to be repeated are to be discussed on a case-by-case basis with the Medical Monitor.

In the case of rescreening, a new participant number is to be assigned. Participants may only participate in the study once.

5.4.2 Withdrawal of Consent and Discontinuation of Participants from the Study

Participants will be informed that they have the right to withdraw consent or discontinue from the study at any time for any reason, without prejudice to their medical care.

Following enrolment into the study, if a participant is discontinued or withdraws consent for any reason, whether related to the investigational product or not, the participant will be considered an early-discontinued participant. The reason for study discontinuation or withdrawal of consent is to be documented in the participants' source documents and electronic Case Report Form (eCRF).

1. Withdrawal of Consent to Participate in the Study

Withdrawal of consent occurs only when a participant (or their substitute decision maker/person responsible) does not want to participate in the study any longer, does not want any further study related visits or assessments and does not want any further study related contact. The Sponsor will continue to retain and use all research results that have already been collected for the study evaluation up until the point of withdrawal of consent. All biological samples that have already been collected may be retained and analyzed at a later date (or as required by local regulations).

If a participant withdraws consent, the Investigator should make an effort to determine the primary reason for this decision (however the participant is not required to provide a reason) and record this information along with the withdrawal of consent discussion, which should be documented in the patient notes. With the participants consent, the Investigator may perform some final safety evaluations to ensure that there are no safety risks associated with the participants withdrawal. Further attempts to contact the participant for study related discussions are not allowed unless safety findings require communication or follow-up.

2. Investigator Decision to Discontinue the Participant

The Investigator also has the right to discontinue participants from the study for any of the following reasons:

- Participant non-adherence to protocol requirements;
- Any other reason based upon the medical judgment of the Investigator.

The Investigator will make every effort to ensure that early-discontinued participants who have received investigational product complete safety follow-up assessments (refer to Schedule of Assessments).

3. Lost to Follow-up.

Reasonable efforts will be made to keep in contact with enrolled study participants. Participants who have not responded to at least 3 such attempts for follow-up may be deemed 'lost to follow-up'. Efforts made to contact study participants must be documented in the participant's file.

5.4.2.1 Individual Stopping Rules

As ^{64}Cu-SAR-bisPSMA will be administered as a bolus IV injection on a single occasion, there are no individual stopping rules.

5.4.3 Pregnancy

There is no information about the effects that ^{64}Cu-SAR-bisPSMA could have on the development of the fetus in humans. Therefore, it is important that female partners of sexually active participants, do not become pregnant during the study and agree to use adequate contraception. Participants will be instructed that known or suspected pregnancy occurring during the study in female partners, should be confirmed and reported to the Investigator.

The Investigator should also be notified of a pregnancy of a female partner of the participant occurring during the study. In the event that a female partner of a participant is found to be pregnant whilst the participant is enrolled in the study, the pregnancy will be followed, and the status of mother and child will be reported to the Sponsor (or delegate).

A Pregnancy Report Form will be completed and submitted for any pregnancy occurring in female partners of participants from enrolment until study completion.

5.4.3.1 Acceptable Forms of Contraception

Sexually active male participants with female partners of childbearing potential are eligible to participate if they agree to follow one of the following methods of contraception consistently, starting from Screening, during the study and for at least 5 days after the last dose of ^{64}Cu-SAR-bisPSMA:

- Are abstinent from penile-vaginal intercourse as their usual and preferred lifestyle (abstinent on a long-term and persistent basis) and agree to remain abstinent;
- Are sterilized (with the appropriate post-vasectomy documentation of the absence of sperm in the ejaculate);
- Agree to use a male condom and have their partner use a contraceptive method with a failure rate of <1% per year as described below when having penile-vaginal intercourse with a woman of childbearing potential who is not currently pregnant, and who agrees to the use of a condom by her partner.

Examples of acceptable contraception include:

- Established use of oral, injected or implanted hormonal methods of contraception;
- Placement of an intrauterine device or intrauterine system.

Examples of non-acceptable methods of contraception include:

- Condoms alone or double barrier;
- Periodic abstinence (e.g., calendar, ovulation, symptothermal, post ovulation);
- Withdrawal;
- Spermicide.

Sexually active male participants with a pregnant or breastfeeding partner must agree to remain abstinent from penile-vaginal intercourse; or use a male condom during each episode of penile penetration during the study.

5.4.3.2 Sperm Donation

Male participants must refrain from donating sperm starting from screening, during the study and for at least 5 days after the administration of ^{64}Cu-SAR-bisPSMA.

5.4.4 Replacement of Participants

Early discontinued participants may be replaced if needed to achieve the target sample size. Participants who withdraw prior to receiving investigational product or prior to undergoing any ^{64}Cu-SAR-bisPSMA PET/CT will be replaced. The Sponsor should be contacted if the site intends to withdraw a patient if feasible, or immediately after withdrawal to discuss the possibility of a replacement.

5.5 Premature Termination or Suspension of Study

The study can be temporarily suspended or prematurely terminated at any time for any reason by the Sponsor if there is sufficient reasonable cause. Written notification, documenting the reason for study suspension or termination, will be provided by the Sponsor to the Investigator, and regulatory authorities. The Investigator may discontinue their study site from participation at any time by way of written notification to the Sponsor indicating their reason for discontinuation.

Should this be necessary, the participants should be seen as soon as possible to perform an End of Study Visit. The Investigator may be informed of additional procedures to be followed in order to ensure that adequate consideration is given to the protection of the participant's interests. As directed by the Sponsor, all study materials must be collected and all eCRFs completed to the greatest extent possible.

If the study is prematurely terminated or suspended, or if the Investigator discontinues their site from the study, the Investigator will be responsible for promptly informing the institutional review board (IRB) and will provide the reason(s) for the termination, suspension or site discontinuation. The Sponsor will be responsible for notifying the appropriate regulatory authorities.

Circumstances that may warrant termination or suspension include, but are not limited to:

- Determination of unexpected, significant, or unacceptable risk to participants;
- Demonstration of efficacy that would warrant stopping;
- Insufficient compliance to protocol requirements;
- Data that are not sufficiently complete and/or evaluable;
- Sponsor terminates the study for administrative, financial or other reasons.

5.6 Safety Review Committee

A Safety Review Committee is not deemed necessary for this study based on what is known with regard to the safety of ^{64}Cu-SAR-bisPSMA from other studies. The Medical Monitor will periodically review laboratory and AE data and will review SAE reports in real time. Any clinically important findings will be shared with the Investigators.

6 INVESTIGATIONAL PRODUCT

Refer to the Pharmacy Manual for further information on supply, storage, dispensing, and disposition of the investigational product.

6.1 Dosage and Formulation

The investigational product administered in this study is ^{64}Cu-labelled SAR-bisPSMA (^{64}Cu-SAR-bisPSMA). The Sponsor is developing this radiopharmaceutical for the diagnosis (PET imaging) of PSMA-expressing PC.

^{64}Cu-SAR-bisPSMA has three basic components: (a) the radionuclide (^{64}Cu) chelated by (b) bisCOSAR (a bifunctional metal chelator) which is linked to (c) two urea based small molecule inhibitors of PSMA.

^{64}Cu-SAR-bisPSMA will be administered as a single bolus IV injection. ^{64}Cu-SAR-bisPSMA will be formulated as a sterile solution for IV injection suitable for human use.

^{64}Cu-SAR-bisPSMA is considered an investigational product by the FDA, and as such will be handled under controlled conditions for use in clinical trials.

^{64}Cu-SAR-bisPSMA is manufactured under relevant regulatory requirements and applicable Good Manufacturing Practice standards.

Details of the investigational product administration are found in Table 4.

Table 4 Investigational Product to be Administered

Investigational Product	Route of administration	Dose (MBq)	Dose Frequency
^{64}Cu-SAR-bisPSMA	Intravenous Single Bolus Injection	200 MBq	Single administration on Day 1

6.2 Supply, Packaging and Labeling

^{64}Cu-SAR-bisPSMA will be manufactured, packaged, and labelled on behalf of the Sponsor and will be provided with appropriate documentation for human use.

The release of each final batch of ^{64}Cu-SAR-bisPSMA for human use is subject to the quality control measures (product specifications) established by the Sponsor, with quality control testing undertaken by the manufacturing facilities. Available analytical data for each batch will be reviewed by qualified personnel at the facility to ensure suitability for human administration.

6.3 Storage and Stability

^{64}Cu-SAR-bisPSMA will be stored in a secure location (i.e., limited access area).

The Investigator (or suitably qualified designee) agrees not to dispense the ^{64}Cu-SAR-bisPSMA from, nor store them at, any site other than the approved study sites as per local regulations.

Once received at the clinical site, ^{64}Cu-SAR-bisPSMA will be stored in a secure location at room temperature with an expiration time designated on the label.

^{64}Cu decays to stable (non-radioactive) daughter products.

6.4 Dispensing and Administration

^{64}Cu-SAR-bisPSMA should be received, used, and administered only by suitably qualified persons at the study center. The receipt, storage, use, transfer, and disposal of ^{64}Cu-SAR-bisPSMA is subject to the regulations and/or appropriate license of the study center.

^{64}Cu-SAR-bisPSMA should be handled by the user in a manner which satisfies both radiation safety and pharmaceutical quality requirements. Appropriate aseptic precautions should be taken.

^{64}Cu-SAR-bisPSMA will be administered as a bolus IV injection under the supervision of the study Investigator or medically qualified delegate at 200 MBq per administration of ^{64}Cu-SAR-bisPSMA.

The Investigator or medically qualified delegate should ensure that the injected radioactivity is within ±10% of the prescribed activity.

6.4.1 Radiation protection

^{64}Cu-SAR-bisPSMA is a radiopharmaceutical, therefore should only be used by personnel who are qualified by training and experience in the safe use and handling of radionuclides and whose experience and training have been approved by the appropriate governmental agency authorized to license the use of radionuclides.

Radiation protection precautions must be taken in accordance with national and local regulations. Full details of restrictions and guidelines to the participant, friends, family and caregivers may be described in the ICF.

6.5 Compliance and Accountability

The Investigator (or designee) is responsible for investigational product accountability, reconciliation, and record maintenance at the investigational site. In accordance with all applicable regulatory requirements, the Investigator or designated site staff must maintain investigational product accountability records throughout the course of the study. The amount of investigational product received from the Sponsor, and the amount administered to participants, and any amount accidentally or deliberately destroyed will be documented.

6.5.1 Storage of Supplies

Study materials will be stored by the study personnel according to the documentation provided with the study materials. The Sponsor reserves the right to inspect the investigational product storage area before and during the study.

6.5.2 Control of Supplies

Dispensing of the investigational product will be carefully recorded on appropriate investigational product accountability forms and will be verified by the study monitor during monitoring visits.

The accountability logs should include dates, quantities, batch numbers, and any unique pack numbers assigned to the investigational product and/or participant. The accountability logs will also include general details related to the study including the protocol, Sponsor and the Investigator.

6.5.3 Disposition of Supplies

The study monitor will review the investigational product accountability logs which detail the disposition of used and unused investigational products by the site.

6.6 Investigational Product Administration

6.6.1 Participant Preparation Prior to Dosing

There is no need for the participant to fast before the administration of ^{64}Cu-SAR-bisPSMA.

Instruct the participant to drink a sufficient amount of water to ensure adequate hydration prior to administration of ^{64}Cu-SAR-bisPSMA and to continue to drink and void frequently following administration to reduce radiation exposure, particularly during the first hour after administration.

6.7 Participant Release and Radioprotection Precautions

Release of participants following administration of ^{64}Cu-SAR-bisPSMA will be in accordance with any applicable regulations at the study center.

7 CONCOMITANT MEDICATIONS

Any medication or vaccine (including over the counter or prescription medicines, vitamins, and/or herbal supplements) that the participant receives during the study must be recorded on the eCRF along with:

- Reason for use;
- Dates of administration including start and end dates;
- Dosage information including dose and frequency.

If the use of any concomitant treatment becomes necessary (e.g., for treatment of an AE), the treatment and administration details must be recorded in the source documents and the eCRF.

7.1 Required Treatments

Upon completion of Visit 4, participants will proceed to RP with PLND (see Section 8.4).

For participants who do not proceed to RP with PLND, obtaining histopathology from excisional biopsy or laparoscopic PLND will be attempted, if this is clinically feasible. This is to be completed prior to the participant undergoing any alternate intervention for PC.

If PLND (either as part of the RP or excisional biopsy or laparoscopic PLND) is not completed, this will be recorded as a protocol deviation.

7.2 Permitted Treatments

All SOC treatments are permitted throughout the duration of a participant's involvement in the study with restrictions and exceptions as outlined in Section 7.3.

7.3 Prohibited Treatments

From enrolment into the study until completion of RP with PLND (or excisional biopsy or laparoscopic PLND), the following interventions are strictly prohibited:

- ADT;
- Any investigational neoadjuvant agent;
- Any therapeutic intervention for PC;
- Any investigational therapy.

Patients who are known to require prohibited treatment should not be considered for participation in the study.

The Medical Monitor **must** be notified of all prohibited medications/treatments administered to any participant, in order to assess the participant's eligibility to continue in the study.

8 STUDY PROCEDURES AND ASSESSMENTS

The following sections describe the study procedures and assessments to be performed during the study. Unplanned visits not specified by the protocol or unscheduled assessments may be performed as clinically indicated at the discretion of the Investigator; the associated data should be recorded on the relevant eCRF in support of an AE diagnosis or lesion assessments. Study procedures and their timing are summarized in the schedule of assessment tables. Protocol waivers or exemptions are not allowed.

Refer to Table 5 for the schedule of assessments for all participants and additional assessments in Table 6 for participants in the PK substudy.

Table 5 Schedule of Assessments – All Participants

	Screening	^{64}Cu-SAR-bisPSMA Imaging			Safety Call (Proceed to RP with PLND after this visit[1])	End of Study Visit (Post-Histopathology Follow-up)[2]
Visit	1	2		3	4	5
Study Day (Window)	Day-28 to 1	Day 1		Day 2	Day 7 (±4 days)	Until Day 90
		Pre Dose	Post Dose			
Informed Consent	X					
Inclusion/Exclusion	X					
Demographics	X					
Body Weight	X	X				
Height	X					
Medical and Medication History	X					
Germline Information (If Available)	X					
AEs & Concomitant Medications	X	X	X	X	X[3]	
Hematology and Biochemistry[4]	X			X[5]		
eGFR (CKD-EPI) in mL/min	X			X[6]		
Biochemical Markers (Total PSA, Testosterone)	X					
Vital Signs (Blood Pressure, Heart Rate)[7]	X	X	X (pre-imaging)	X		
Duplicate 12-Lead ECG		X	X[8]			
Urinalysis[9]	X			X[10] (pre-imaging)		
Institutional SOC Imaging[11]	X					
^{64}Cu-SAR-bisPSMA Dose		X				
^{64}Cu-SAR-bisPSMA PET/CT[12]			X (1 to 4h post dose)	X (24 ±6h post dose)		
Histopathology Results	X (primary prostate)					X (prostate, pelvic LNs and distant metastasis)
RP and PLND Details						X
Alternate PC Intervention (If Applicable)[13]						X

Abbreviations: AE=adverse event; CKD-EPI=Chronic Kidney Disease Epidemiology Collaboration; CT=computed tomography; eGFR=estimated glomerular filtration rate; ECG=electrocardiogram; PC=prostate cancer; PET=positron emission tomography; PLND=pelvic lymph node dissection; PSA=prostate-specific antigen; RP=radical prostatectomy; SOC=

standard of care

[1] Upon completion of Visit 4, participants will proceed to RP with PLND (see Section 8.4). For participants who do not proceed to RP with PLND, obtaining histopathology from excisional biopsy or laparoscopic PLND will be attempted, if this is clinically feasible. This is to be completed prior to the participant undergoing any alternate intervention for PC.
[2] Visit 5 is to be completed after the participant has undergone RP with PLND (or excisional biopsy or laparoscopic PLND) and the histopathology results for pelvic LNs (prostate and distant metastasis, if appliable) are available. Visit 5 may be completed via a telephone call.
[3] If an AE is experienced, additional safety laboratory assessments may be completed as an unscheduled visit.
[4] Hematology: hemoglobin, hematocrit, red blood cell count, white blood cell count, platelet count, absolute neutrophil count, and absolute lymphocyte count. Biochemistry: albumin, total protein, random blood glucose, sodium, potassium, blood urea nitrogen or urea, creatinine, calcium, uric acid, aspartate aminotransferase, alanine aminotransferase, and total bilirubin.
[5] To be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration.
[6] To be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration. For the calculation of Body Surface Area, height as assessed at Screening, and weight as assessed on Day 1 is to be used.
[7] To be performed after the participant has rested comfortably in a seated position with feet on the floor or in a supine position for at least 3 minutes.
[8] Post-dose ECG to be performed at 30 minutes (±10 minutes) post ^{64}Cu-SAR-bisPSMA administration.
[9] Urinalysis to include determination of protein, glucose and leukocytes (dipstick test).
[10] To be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration. If urinalysis is completed during the Day 2 visit, it is to be completed prior to the PET/CT.
[11] Institutional SOC imaging is defined as any single modality or a combination of modalities allowing the assessment of visceral organs, soft tissue, and bone in the chest, abdomen and pelvis (e.g., MRI, CT, approved SOC PSMA PET/CT, with or without ^{99m}Tc-MDP bone scan). If the required scans were performed within 60 days of Day 1, these scans will serve as baseline SOC imaging and do not need to be repeated at screening. If the required scans are not available, the participant will undergo the missing scan(s) during the Screening period. A washout period of at least 72 hours must be applied between the administration of approved PSMA PET tracers and ^{64}Cu-SAR-bisPSMA.
[12] A whole-body low dose CT and PET scan will be acquired from the mid-thigh to the skull vertex. Participants to void their bladder immediately before the acquisition of the PET/CT scan.
[13] If a participant's planned procedure (RP with PLND) for PC changes, the alternate PC intervention and reason for the change will be recorded in the eCRF.

Table 6 Schedule of Assessments – PK Substudy Participants Only

Study Day	**Day 1**								**Day 2**	**Day 3**
Time (window)	**Pre Dose (T -1h to T0)**	**T0[1] to T5min**	**T0 +10min (±2min)**	**T0 +30min (±5min)**	**T0 +1h (±15min)**	**T0 +2h (±15min)**	**T0 +4h (±30min)**	**T0 +6h (±30min)**	**T0 +24h (±2h)**	**T0 +48h (±2h)**
Blood & Plasma sample[2]	X	X	X	X	X	X	X	X	X	X
Urine collection[3]		X (T0 to T2h)				X (T2h to T4h)	X (T4h to T6h)			
^{64}Cu-SAR-bisPSMA PET/CT[4]					X		X	X	X	X
AEs & Concomitant Medications										X

Abbreviations: AE=adverse events; CT=computed tomography; h=hour; min=minute; PET=positron emission tomography; T=time

[1] T0 is the time of ^{64}Cu-SAR-bisPSMA administration.
[2] Blood sampling is to be completed before the PET/CT acquisition (where applicable). See Section 8.7.1 for details.
[3] Urine samples will be collected over the following intervals: 0 to 2h, 2 to 4h, 4 to 6h post injection of ^{64}Cu-SAR-bisPSMA. Participants to void their bladder at the end of the collection windows. See Section 8.7.1 for details.
[4] Participants to void their bladder immediately before the acquisition of the PET/CT scan.

8.1 Demographics and Baseline Health Record

8.1.1 Baseline Demographics

Participant demographics will be recorded, including year of birth, race, ethnicity.

8.1.2 Body Weight and Height

Body weight (in kg) (wearing light clothes, no shoes) and height (in cm) will be measured (rounded to one decimal place).

8.1.3 Medical and Medication History

A complete prostate medical history will be obtained at screening and the following disease characteristics will be recorded, including date of initial diagnosis, tumor type, Gleason Score, International Society of Urological Pathology (ISUP) Grade Group, number of positive cores and locations of positive cores, clinical TNM staging and risk category (high or very high risk).

Other clinically relevant medical history will also be collected. This should include any past and/or current medical conditions that, in the opinion of the Investigator, are clinically relevant regardless of whether it has been resolved, any prior malignancies other than PC or concurrent or past conditions that can change pelvic anatomy or cause lymphadenopathies (e.g., viral, bacterial or parasitic infections).

Medication history (including vaccine, over the counter or prescription medicines, vitamins, and/or herbal supplements) that the participant is receiving within 14 days before signing informed consent must be recorded on the eCRF along with:

- Reason for use;
- Dates of administration including start and end dates;
- Dosage information including dose and frequency.

8.1.4 Germline Information

If available, germline information will be collected, including date of test and result (pathogenic germline variant, variant of uncertain clinical significance or benign). Details of any specific gene variant will also be collected where specified.

8.2 Safety Assessments

8.2.1 Vital Signs

Vital signs measurements will include systolic and diastolic blood pressure and heart rate. Wherever possible, vital signs will be recorded after the participant has rested comfortably in a seated position with feet on the floor or in a supine position for at least 3 minutes (the position must be the same for a particular participant pre and post dose) and using consistent methods between participants.

Isolated clinically significant observations, as deemed by the Investigator, will be recorded as AEs.

8.2.2 Duplicate 12-lead Electrocardiogram

Duplicate 12-lead ECGs will be performed after the participant has rested comfortably in the supine position for at least 3 minutes. The following parameters will be assessed and recorded on the eCRF: heart rate, PR, RR, QRS, QT, QTcB (Bazett formula), QTcF (Fridericia formula). The Investigator (or a qualified delegate at the investigational site) will interpret the ECG using one of the following categories: within normal limits, abnormal but not clinically significant, or abnormal and clinically significant. Abnormal findings will also be specified on the eCRF.

Clinically significant observations, as deemed by the Investigator, will be recorded as AEs.

8.2.3 Clinical Laboratory Assessments

Clinical laboratory samples will be analyzed by the site's local laboratory.

Blood samples will be collected for the following tests:

- Hematology: Hemoglobin, hematocrit, red blood cell count, white blood cell count, platelet count, absolute neutrophil count, and absolute lymphocyte count.
- Biochemistry: albumin, total protein, random blood glucose, sodium, potassium, blood urea nitrogen or urea, creatinine, calcium, uric acid, aspartate aminotransferase, alanine aminotransferase, and total bilirubin.

  Time of last meal is to be recorded on the eCRF.
- Estimated glomerular filtration rate (eGFR): will be calculated from creatinine using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula and reported as mL/min. To convert mL/min/1.73m^2 to mL/min, multiply by the individual's Body Surface Area (calculated using the Mosteller formula) and divide by 1.73.
- Biochemical markers: total PSA and testosterone.

Urine samples will be collected for the following tests:

- Urinalysis: to include determination of protein, glucose and leukocytes (dipstick test).

In the event of an unexplained, clinically significant abnormal laboratory test result, the test should be repeated and followed up until it has returned to the normal range and/or an adequate explanation of the abnormality is found. During the study, all out of range (abnormal) laboratory values (except for total PSA and testosterone), must be evaluated and commented on by the Investigator (or a qualified delegated observer at the investigational site) for clinical significance using one of the following categories: within normal limits, abnormal but not clinically significant, or abnormal and clinically significant. Clinically significant observations will be recorded as AEs unless they are associated with a diagnosis that itself has been reported as an AE (e.g., raised white cells and fever associated with an infection – in this case the infection is the AE, and the lab value and fever are signs and symptoms of the AE and should not be reported separately).

8.3 Imaging Assessments

For a given participant, the same PET scanner is preferred to be used to acquire the approved SOC PSMA PET/CT (if this scan is part of the institutional SOC imaging work up at screening), and the Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT. If this is not possible, a scanner of the same make and model is requested. If neither are possible any SOC PSMA PET

or other SOC scan as per each institutional standard is accepted. The Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CTs are required to be done on the same scanner.

8.3.1 Institutional SOC Imaging

Institutional SOC imaging will be performed as per the clinical study site's standard procedures. All participants are required to have baseline SOC imaging which is defined as any single modality or a combination of modalities allowing the assessment of visceral organs, soft tissue, and bone in the chest, abdomen and pelvis (e.g., MRI, CT, approved SOC PSMA PET/CT, with or without ^{99m}Tc-MDP bone scan). If the required scans were performed within 60 days of Day 1, these scans will serve as baseline SOC imaging and do not need to be repeated at screening. If the required scans are not available, the participant will undergo the missing scan(s) during the Screening period.

A washout period of at least 72 hours must be applied between the administration of approved PSMA PET tracers and ^{64}Cu-SAR-bisPSMA.

The imaging modality, imaging type, date of acquisition, tracer used (if applicable), contrast used (if applicable), and the local interpretation for the relevant regions will be recorded on the eCRF.

For the central image analysis of the institutional SOC imaging, the site will transfer the images in a Digital Imaging and Communications in Medicine (DICOM) format to the central imaging core lab with de-identification of participant specific information as specified in the protocol-specific site imaging manual.

8.3.1.1 Institutional SOC Imaging Central Interpretation

An independent imaging core lab will receive and evaluate the SOC images. The scans will be interpreted by a single independent, blinded, central reader without access to the ^{64}Cu-SAR-bisPSMA PET/CT scans (refer to Section 5.3 for further detail on blinding). The reader will assess the same regions/subregions as described in Table 7 and record the number of lesion(s). The size, SUV and TBR will be recorded for up to 5 lesions per region.

For details on the interpretation of the SOC imaging refer to the Imaging Review Charter.

Note: The result of the central review of the study scans will not be provided to the clinical site.

8.3.2 ^{64}Cu-SAR-bisPSMA PET/CT

The site will conduct the ^{64}Cu-SAR-bisPSMA PET/CT scans according to the protocol-specific site imaging manual, which contains details of scan acquisition, processing parameters and submission procedures. Participants will be asked to void immediately prior to each scan acquisition. After voiding, a whole-body low dose CT and PET scan will be acquired from the mid-thigh to the skull vertex. The date, start of scan acquisition, and scanner used will be recorded on the eCRF.

All ^{64}Cu-SAR-bisPSMA PET/CT scans will be transferred in DICOM format to the central imaging core lab with anonymization of participant specific information as specified in the protocol-specific site imaging manual.

8.3.2.1 ^{64}Cu-SAR-bisPSMA PET/CT Local Interpretation

The Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT scans will be interpreted locally by an appropriately qualified Investigator in a non-blinded fashion. The reader will be required to evaluate the scans individually for the presence of abnormal ^{64}Cu-SAR-bisPSMA uptake in the pelvic LNs, prostate gland, extra-pelvic LNs, visceral/soft tissue and bone. A positive lesion will be defined as uptake of ^{64}Cu-SAR-bisPSMA that is greater than local background activity and judged by the reader to be consistent with PC. The number of PET-positive lesion(s) in each subregion will be documented (See Table 7). CT scans from the ^{64}Cu-SAR-bisPSMA PET/CT will be available to the reader to guide anatomical localization.

Table 7 Regions and Subregions Assessed

Regions:	**Subregion(s):**
1. Prostate	Gland (including seminal vesicles)
2. Pelvic LN	Right Pelvic, Left Pelvic, Other
3. Extra-Pelvic LN	Retroperitoneal, Supradiaphragmatic, Other
4. Bone	Pelvic Bone Right, Pelvic Bone Left, Skull, Clavicle Right, Clavicle Left, Scapula Right, Scapula Left, Sternum, Ribcage Right, Ribcage Left, Spine, Extremities, Other
5. Visceral/soft tissue	Lung/pleura, Liver, Other

If distant metastatic disease (lesion[s] within the extra-pelvic LNs, bone or visceral/soft tissue regions) is detected on the ^{64}Cu-SAR-bisPSMA PET/CT, histopathology (from image-guided biopsy or surgery) should be obtained if clinically feasible (see Section 8.5.2). The specimens will be processed and analyzed locally.

Any incidental findings on the ^{64}Cu-SAR-bisPSMA PET/CT scan(s) that are of potential clinical relevance and are not directly associated with the objectives of the protocol should be evaluated and handled by the Investigator per standard medical/clinical judgment.

8.3.2.2 ^{64}Cu-SAR-bisPSMA PET/CT Central Interpretation

An independent imaging core lab will receive and evaluate the ^{64}Cu-SAR-bisPSMA PET/CT scans. The Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT scans will be interpreted by three different independent blinded readers in a random order at separate reading sessions. Central readers will be blinded as described in Section 5.3. The readers will be required to evaluate the Day 1 and Day 2 scans individually for the presence of abnormal ^{64}Cu-SAR-bisPSMA uptake in the pelvic LNs, prostate gland, extra-pelvic LNs, visceral/soft tissue and bone. The same definition of a ^{64}Cu-SAR-bisPSMA PET-positive lesion will be applied as for the local interpretation and the number of PET-positive lesion(s) in each subregion will be documented (refer to Section 8.3.2.1 for the definition of a PET-positive lesion and list of regions and subregions). A subregion/region will be considered positive for PC if it contains at least one PET-positive lesion and negative for PC if no lesion is recorded. CT scans from the ^{64}Cu-SAR-bisPSMA PET/CT will be available to the reader to guide anatomical localization.

Each central reader will also record the size, SUV, TBR and reader confidence level for up to 5 lesions per region on each Day 1 and Day 2 scan.

For the process of image randomization and interpretation refer to the Imaging Review Charter.

Note: The result of the central review of the study scans will not be provided to the clinical site.

8.4 Radical Prostatectomy and Pelvic Lymph Node Dissection

Upon completion of Visit 4, participants will undergo RP with PLND as detailed in the Surgery Manual. Surgical staff will label all specimens removed during surgery to correlate with the specific anatomic location of removal (including region and subregion). The specimens will be processed and analyzed locally.

If a participant does not proceed to RP with PLND, obtaining histopathology from excisional biopsy or laparoscopic PLND will be attempted, if this is clinically feasible. This is to be completed prior to the participant undergoing any alternate intervention for PC.

The procedure, date of surgery and extent of PLND will be captured in the eCRF.

If alternate intervention for PC is completed, the date, type of treatment and reason for the change in management will be recorded.

8.5 Histopathology

8.5.1 Tissue from Radical Prostatectomy and Pelvic Lymph Node Dissection

Immediately following the RP with PLND (or excisional biopsy or laparoscopic PLND as per Section 8.4), the labelled specimens will be sent for histopathological processing and analysis at the site in accordance with the Pathology Manual. The local pathologist evaluating the samples will be blinded to the ^{64}Cu-SAR-bisPSMA PET/CT scans/results.

The date of assessment and histopathology result will be extracted from the local report and recorded on the eCRF within 90 days of Day 1 (Visit 5). Refer to the Pathology Manual for reporting requirements.

8.5.2 Tissue from Distant Metastasis

If distant metastatic disease (lesion[s] within the extra-pelvic LNs, bone or visceral/soft tissue regions) is detected on the ^{64}Cu-SAR-bisPSMA PET/CT per the local interpretation of the scans, histopathology (from image-guided biopsy or surgery) should be obtained if clinically feasible. The specimens will be processed and analyzed locally according to the institutional protocols.

For each distant metastatic lesion with histopathology, the procedure, date, imaging modality used to guide biopsy (if applicable), region and subregion allocation, size and local histopathology result for PC will be recorded on the eCRF within 90 days of Day 1 (Visit 5).

8.6 Efficacy Assessments

To determine the efficacy of ^{64}Cu-SAR-bisPSMA PET/CT at Day 1 and Day 2, the scan result on the respective day will be matched against the Standard of Truth result. Matching of the Standard of Truth against the ^{64}Cu-SAR-bisPSMA PET/CT result will be completed via statistical programming.

8.6.1 Standard of Truth for Pelvic Lymph Nodes

The Standard of Truth for the pelvic LN subregions will be based on histopathology from PLND completed as part of the RP (or excisional biopsy or laparoscopic PLND as per Section 8.4). The Standard of Truth for a given subregion will be deemed ***positive*** if it contains at least one pelvic LN that is positive for PC based on histology, ***negative*** if PLND was completed but

no pelvic LN was positive for PC and ***non-evaluable*** if the histopathology data are either not available or deemed insufficient for the evaluation of pelvic LN involvement.

8.6.2 Standard of Truth for Prostate

According to the inclusion criteria, all enrolled participants must have a positive diagnosis of primary PC confirmed by histopathology at study entry. Therefore, the Standard of Truth for the **prostate region** will automatically be assigned as ***positive***.

8.6.3 Standard of Truth for Distant Metastatic Disease

Where clinically feasible, histopathology should be obtained for distant metastatic lesion(s) detected on the ^{64}Cu-SAR-bisPSMA PET/CT based on the local interpretation of the scans (refer to Sections 8.3.2.1 and Section 8.5.2).

The Standard of Truth for distant metastasis (**extra-pelvic LNs, bone and visceral/soft tissue regions**) will be deemed ***positive***, if at least one distant metastatic lesion is positive for PC based on histology, ***negative*** if histopathology was obtained, but no lesion was positive for PC and ***non-evaluable*** if histopathology data are either not available or deemed insufficient for the evaluation of distant metastatic PC.

8.7 Assessment of Pharmacokinetics

A subset of 10 participants enrolled into the PK substudy will complete additional assessments on Day 1, Day 2, and Day 3 as per Table 6. For detailed instructions on PK assessments, please refer to the PK Manual.

8.7.1 Blood and Urine Pharmacokinetics

Aliquots of blood, plasma and urine samples will be analyzed to determine radioactivity counts using a gamma well counting method immediately following the sample collection period. The resulting radioactivity counts will be converted into radioactivity concentrations using a reference standard.

Blood concentration-time data will be analyzed with non-compartmental analysis methods and pharmacokinetic parameters will be generated. The blood PK parameters to be estimated are described in Section 11.4.13. Ratios of blood-to-plasma radioactivity at each time point will be calculated.

The total amount of excreted radioactivity per urine collection interval will be calculated as a product of urine radioactivity concentration and total volume collected over the time interval, and then expressed as a percent of administered dose. Urine concentration-time data will be analyzed with non-compartmental analysis methods and pharmacokinetic parameters will be generated. The urine PK parameters to be estimated are described in Section 11.4.13.

Metabolic profiles in urine and plasma will be determined using radio-chromatographic methods. The areas under the radio-chromatographic peaks corresponding to the parent compounds and putative metabolites, if applicable, will be used in the quantitative assessment of metabolic profiles in urine and plasma.

For detailed instructions on sample collection, processing, and analysis, please refer to the PK Manual.

8.7.2 Radiation Dosimetry

The radiation dosimetry of ^{64}Cu-SAR-bisPSMA for the organ/tissue absorbed doses (as mGy/MBq) and whole body effective dose (as mSv/MBq and mSv/200 MBq) will be determined using standard methods established by the Committee on Medical Internal Radiation Dose.

For participants in the PK substudy, whole body PET/CT images will be acquired at 1, 4, 6, 24 and 48 hours post injection of ^{64}Cu-SAR-bisPSMA (note: the number of scans and acquisition timepoints for the assessment of dosimetry may be modified during the study based on emerging data at the Sponsor's discretion). The PET/CT images will be used to obtain volumes and activity concentrations for selected tissues/organs. Based on these data, time-integrated activity coefficients will be derived and used for subsequent calculation of radiation absorbed doses using the OLINDA/EXM software. The assessment of dosimetry will be completed centrally.

9 VISIT-SPECIFIC PROCEDURES

9.1.1 Screening Period

9.1.1.1 Visit 1 (Day -28 to Day 1)

The following assessments must be performed/obtained within the 28 days before the participant receives the ^{64}Cu-SAR-bisPSMA administration. Informed consent must be documented before any study-specific screening procedure is performed and may be obtained more than 28 days before the ^{64}Cu-SAR-bisPSMA administration.

Results of SOC tests or examinations performed prior to obtaining informed consent and within 28 days prior to the ^{64}Cu-SAR-bisPSMA administration (60 days in case of the institutional SOC scans) may be used; such tests do not need to be repeated for screening.

If the start date for the ^{64}Cu-SAR-bisPSMA administration extends beyond 28 days from the start of the screening assessments (e.g., due to issues related to logistics), the Investigator in consultation with the Medical Monitor must determine if any screening assessments need to be repeated.

Assessments to be completed:

- Informed Consent
- Inclusion/Exclusion
- Demographics
- Body Weight and Height
- Medical and Medication History
- Germline Information (If Available)
- Adverse Events (ongoing)
- Concomitant Medications (ongoing)
- Hematology and Biochemistry
- eGFR (CKD-EPI) in mL/min
- Biochemical Markers (Total PSA, Testosterone)
- Vital Signs (Blood Pressure, Heart Rate)
- Urinalysis
- Institutional SOC imaging (e.g., MRI, CT, approved SOC PSMA PET/CT, with or without ^{99m}Tc-MDP bone scan)
- Histopathology Results (primary prostate)

9.1.2 ^{64}Cu-SAR-bisPSMA Imaging

9.1.2.1 Visit 2 (Day 1)

Day 1 = Day of ^{64}Cu-SAR-bisPSMA administration.

A washout period of at least 72 hours must be applied between the administration of approved PSMA PET tracers (if this is the institutional SOC imaging at screening) and ^{64}Cu-SAR-bisPSMA. Pre dose assessments to be completed on the day of but prior to the ^{64}Cu-SAR-bisPSMA administration. Post dose assessments to be completed as per the instructions / windows below.

Assessments to be completed:

- Adverse Events (ongoing)

- Concomitant Medications (ongoing)

Pre dose

- Body Weight
- Vital Signs (Blood Pressure, Heart Rate)
- Duplicate 12-Lead ECG

^{64}Cu-SAR-bisPSMA Dose

Post dose

- Duplicate 12-Lead ECG (to be performed at 30 minutes [±10 minutes] post ^{64}Cu-SAR-bisPSMA administration)
- Vital Signs (Blood Pressure, Heart Rate) prior to the PET/CT
- ^{64}Cu-SAR-bisPSMA PET/CT at 1 to 4 hours post dose

9.1.2.2 Visit 3 (Day 2)

This visit is to be completed 24 ±6 hours after the ^{64}Cu-SAR-bisPSMA administration (note: urinalysis and laboratory assessments are to be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration).

Assessments to be completed:

- Adverse Events (ongoing)
- Concomitant Medications (ongoing)
- Hematology and Biochemistry (to be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration)
- eGFR (CKD-EPI) in mL/min (to be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration)
- Vital Signs (Blood Pressure, Heart Rate)
- Urinalysis (to be completed within 9 days after the ^{64}Cu-SAR-bisPSMA administration; if urinalysis is completed during the Day 2 visit, it is to be completed prior to the PET/CT)
- ^{64}Cu-SAR-bisPSMA PET/CT at 24 ±6 hours post dose

9.1.3 Safety Call - Visit 4 (Day 7 ±4 days)

The site will contact the participant via telephone between Day 3 to Day 10.

Assessments to be completed:

- Adverse Events (ongoing)
- Concomitant Medications (ongoing)

Upon completion of Visit 4, participants will proceed to RP with PLND (see Section 8.4). For participants who do not proceed to RP with PLND, obtaining histopathology from excisional biopsy or laparoscopic PLND will be attempted, if this is clinically feasible. This is to be completed prior to the participant undergoing any alternate intervention for PC.

9.1.4 End Of Study - Visit 5 (Until Day 90)

This visit is to take place by Day 90 after the participant has completed the RP with PLND (or excisional biopsy or laparoscopic PLND) and the histopathology results for pelvic LNs (prostate and distant metastasis, if appliable) are available. This visit may be completed via a telephone call. The End of Study Visit must take place by Day 90 for all participants, regardless of whether the participant completed the RP/PLND or not.

Assessments to be completed:

- Histopathology Results (prostate, pelvic LNs and distant metastasis)
- RP and PLND Details
- Alternate PC Intervention (if applicable)

10 ADVERSE EVENT MONITORING

10.1 Adverse Events

An AE is defined as any untoward medical occurrence in a clinical investigation participant administered a pharmaceutical product, which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavorable and/or unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of an investigational product, whether or not related to the investigational product. All AEs, including observed or volunteered problems, complaints, or symptoms, are to be recorded on the appropriate eCRF.

AEs should be recorded from the time of informed consent until completion of Visit 4. AEs recorded before first administration of ^{64}Cu-SAR-bisPSMA will be classified as non-treatment emergent AEs. All AEs recorded after ^{64}Cu-SAR-bisPSMA administration will be classified as treatment emergent AEs (TEAEs). Non-related AEs will only be reported and followed up until completion of Visit 4. All related AEs are to be followed up until resolved or judged to be no longer clinically significant, or until they become chronic to the extent that they can be fully characterized. An assessment should be made at the last study-related visit for each patient. Certain long-term AEs cannot be followed until resolution within the settings of this protocol. In these cases, follow-up will be the responsibility of the treating physician. If during AE follow-up the case has progressed to the level of SAE, or if a new SAE is observed whose relationship to the study medication cannot be ruled out, the situation must be reported by the Investigator becoming aware of the information.

Participants should be instructed to report any AE that they experience to the Investigator.

Wherever possible, a specific disease or syndrome rather than individual associated signs and symptoms should be identified by the Investigator and recorded on the eCRF. However, if an observed or reported sign or symptom is not considered a component of a specific disease or syndrome by the Investigator, it should be recorded as a separate AE on the eCRF. Additionally, the condition that led to a medical or surgical procedure (e.g., surgery, endoscopy, tooth extraction, transfusion) should be recorded as an AE, not the procedure. AEs should be entered into the eCRF within 5 days of reporting.

Clinically significant abnormal laboratory or other examination (e.g., ECG) findings that are detected during the study should be reported as AEs except as part of an alternate diagnosis or syndrome. The Investigator will exercise his or her medical and scientific judgment in deciding whether an abnormal laboratory finding, or other abnormal assessment is clinically significant. Clinically significant abnormal laboratory values occurring during the clinical study will be followed until repeat tests return to normal, stabilize, or are no longer clinically significant. Any abnormal test that is determined to be an error does not require reporting as an AE.

Any event that in the opinion of the Investigator is solely due to the underlying disease should not be reported as an AE.

10.1.1 Adverse Drug Reaction

All noxious and unintended responses to a medicinal product related to any dose should be considered an adverse drug reaction. “Responses” to a medicinal product means that a causal relationship between a medicinal product and an AE is at least a reasonable possibility; i.e., the relationship cannot be ruled out.

10.1.2 Unexpected Adverse Drug Reaction

An unexpected adverse drug reaction is defined as an adverse reaction, the nature or severity of which is not consistent with the applicable product information. For ^{64}Cu-SAR-bisPSMA, the reference safety information is included in the current IB. The reference safety information is reviewed yearly, and the periodicity of the review will be harmonized with the reporting period of the Development Safety Update Report.

10.1.3 Assessment of Adverse Events by the Investigator

The Investigator will assess the severity (intensity) of each AE according to National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) criteria and will also categorize each AE as to its potential relationship to investigational product using the categories of related or not related.

10.1.3.1 Assessment of Severity

Severity of AEs will be graded according to definitions below.

The Investigator will make an assessment of severity (intensity) for each AE and SAE reported during the study using the NCI CTCAE V5.0 criteria. The assessments will be based on the Investigator's clinical judgment. The severity of each AE and SAE recorded in the eCRF should be assigned to 1 of the following:

Mild (Grade 1):	Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
Moderate (Grade 2):	Moderate; minimal, local or noninvasive intervention indicated; limiting age appropriate instrumental Activities of Daily Living (ADL)*.
Severe (Grade 3):	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL**.
Life-threatening (Grade 4):	Life-threatening consequences; urgent intervention indicated.
Fatal (Grade 5):	Death related to AE.

*Instrumental ADL refers to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

**Self-care ADL refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not being bedridden.

10.1.3.2 Causality Assessment

The Investigator is obligated to assess the relationship between the investigational product and the occurrence of each AE/SAE. The Investigator will use clinical judgment to determine the relationship to ^{64}Cu-SAR-bisPSMA. Alternative causes, such as natural history of the underlying diseases, concomitant therapy, other risk factors, and the temporal relationship of the event to the investigational product will be considered and investigated.

The causal relationship of the AE to the investigational product or study procedures should be assessed by the Investigator (or medically qualified delegate) using the following classifications:

Related:	An AE that follows a temporal sequence from administration of the investigational product, or for which possible involvement of the investigational product cannot be ruled out, although factors other than the drug, such as underlying diseases, concurrent medical conditions, concomitant drugs and concurrent treatments, are also possibly responsible.
Not related:	An AE that does not follow a temporal sequence from investigational product administration and/or that can reasonably be explained by other factors, such as underlying diseases, concurrent medical conditions, concomitant drugs and concurrent treatments.

10.2 Serious Adverse Event

Any AE, due solely to the underlying disease or disease progression, should not be reported as an SAE, unless the study treatment contributed to the AE.

An AE or adverse reaction is considered serious if, in the view of either the Investigator or Sponsor, it results in any of the following outcomes:

- Death;
- Life-threatening AE;

  Note: An AE or adverse reaction is considered "life-threatening" if, in the view of the Investigator or Sponsor, its occurrence places the participant at immediate risk of death. It does not include an event that, had it occurred in a more severe form, might have caused death.)

- Requires hospitalization or prolongation of existing hospitalization;

  Note: Any hospital admission with at least 1 overnight stay will be considered an inpatient hospitalization. An emergency room visit without hospital admission will not be recorded as an SAE under this criterion, nor will hospitalization for a procedure scheduled or planned before signing of informed consent. However, unexpected complications and/or prolongation of hospitalization that occur during elective surgery should be recorded as AEs and assessed for seriousness. Admission to the hospital for social or situational reasons (e.g., no place to stay, live too far away to come for hospital visits) will not be considered inpatient hospitalizations.

- A persistent or significant disability/incapacity or substantial disruption of the ability to conduct normal life functions;
- A congenital anomaly/birth defect;
- An important medical event;

  Note: Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered an SAE when, based upon appropriate medical judgment, they may jeopardize the participant and may require medical or surgical intervention to prevent 1 of the outcomes listed above. Examples of such

medical events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse.

10.3 Serious Adverse Event Reporting – Procedures for Investigators

10.3.1 Initial Reports

All SAEs occurring from the time of consent until completion of Visit 4 must be reported within 24 hours of the knowledge of the occurrence (this refers to any AE that meets any of the aforementioned serious criteria). All SAEs that the Investigator considers related to investigational product occurring until completion of Visit 4 must be reported to the Sponsor.

To report an SAE, complete the SAE Report Form and email the completed SAE Report Form within 24 hours of awareness. An SAE Report Form will be supplied with the study binder.

In addition to reporting the SAE to Sponsor (or designee), the Investigator must also notify the IRB which approved the study according to their requirements. Copies of all correspondence relating to reporting of any SAEs should be maintained in the site's study files and will be checked routinely by the study monitor.

If the Sponsor and the Investigator consider that the SAE is investigational product related (i.e., an adverse reaction) and unexpected it will be reported to the appropriate regulatory authorities by the Sponsor (or designee) within the pre-defined timelines.

10.3.2 Follow-Up Reports

The Investigator must continue to follow the participant until the SAE has subsided or until the condition becomes chronic in nature, stabilizes (in the case of persistent impairment), or the participant dies.

Within 24 hours of receipt of follow-up information, the Investigator must update the SAE form and submit any supporting documentation (e.g., deidentified participant discharge summary, autopsy reports), if requested.

10.4 Pregnancy

Pregnancies occurring in a female partner of a participant after the first administration of ^{64}Cu-SAR-bisPSMA require immediate reporting. They must be reported within 24 hours after the Investigator has become aware of the pregnancy. A pregnancy report will be completed and sent to the Sponsor (or designee) within 24 hours of becoming aware of the pregnancy. The Investigator will collect follow-up information regarding the outcome of the pregnancy and any postnatal sequelae in the infant that must also be reported to the Sponsor or designee. Upon awareness of the outcome of the pregnancy, the Investigator must forward a follow-up Pregnancy Report with any relevant information.

If the outcome of the pregnancy meets the criteria for immediate classification of an SAE (e.g., spontaneous abortion, stillbirth, neonatal death, or congenital anomaly), the Investigator will report the event within 24 hours of being notified of the pregnancy report.

10.5 Expedited Reporting

The Sponsor will report all relevant information about suspected unexpected serious adverse reactions that are fatal or life-threatening as soon as possible to the FDA and the IRBs, and in any case no later than 7 calendar days after knowledge by the Sponsor of such a case, and

relevant follow-up information will subsequently be communicated within an additional 8 days.

All other suspected unexpected serious adverse reactions will be reported to the FDA and the IRBs as soon as possible but within a maximum of 15 calendar days of first knowledge by the Sponsor.

For non-US sites, the Sponsor will report all suspected unexpected serious adverse reactions according to relevant laws and regulations applicable to the countries / jurisdiction.

The Sponsor will also inform all Investigators as required.

11 STATISTICAL METHODS

11.1 General Considerations

A Statistical Analysis Plan (SAP) will be written after finalizing the protocol and prior to database lock. The SAP will detail the implementation of all the planned statistical analysis in accordance with the principal features stated in the protocol. Any deviations from the SAP will be presented in the final clinical study report.

Continuous variables will be summarized by the number of subjects, mean, standard deviation, standard error of the mean, median, 25th quartile, 75th quartile, minimum, and maximum values. For the summary of PK data, the coefficient of variation as a percentage (CV%), geometric mean and geometric CV% will also be presented where appropriate.

Categorical variables will be summarized using frequency counts and percentages.

For efficacy variables 95% two-sided confidence intervals (CI) will be presented along with 97.5% CIs for the primary endpoint efficacy outcomes.

11.2 Analysis Populations

11.2.1 All Participants Set

The All Participants Set will include all participants that signed informed consent, including those participants that failed screening. The All Participants Set will be used for the listing and summary of screening information collected.

11.2.2 Enrolled Participants Set

The Enrolled Participants Set will include all participants who signed informed consent and were enrolled into the study (regardless of dosing or assessments completed). The Enrolled Participant Set will be used for the summary of participant disposition data.

11.2.3 Evaluable Set

The Evaluable Set will consist of all participants who have a ^{64}Cu-SAR-bisPSMA PET result (positive or negative) and Standard of Truth (positive or negative). The Evaluable Set will be used for the supportive analysis for the primary efficacy endpoints and other selected secondary and exploratory analyses. For analyses conducted per ^{64}Cu-SAR-bisPSMA imaging day, only participants in the Evaluable Set that have a ^{64}Cu-SAR-bisPSMA PET for the respective imaging day will contribute data to the assessment for that imaging day.

11.2.4 Full Analysis Set

The Full Analysis Set will consist of all participants who have at least one ^{64}Cu-SAR-bisPSMA PET result. The Full Analysis Set will be used for the primary efficacy endpoints and other select secondary and exploratory analyses.

11.2.5 Safety Set

The Safety Set will include all participants who receive any amount of ^{64}Cu-SAR-bisPSMA. The Safety Set will be used to assess all safety data.

11.2.6 Blood/Plasma Pharmacokinetic Set

The Blood/Plasma PK Set will include all participants who receive any amount of ^{64}Cu-SAR-bisPSMA and have sufficient blood and plasma concentration data for the determination of at least one PK parameter.

11.2.7 Urine Pharmacokinetic Set

The Urine PK Set will include all participants who receive any amount of ^{64}Cu-SAR-bisPSMA and have urine data collected and analyzed for at least one full urine sampling collection period.

11.2.8 Dosimetry Set

The Dosimetry Set will include all participants who receive any amount of ^{64}Cu-SAR-bisPSMA and complete the ^{64}Cu-SAR-bisPSMA PET/CT scans required for the assessment of dosimetry.

11.3 Determination of Sample Size

The sample size is calculated based on the co-primary endpoints of sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth, on both Day 1 and Day 2 (independently). At least two of the three central readers must reject the null hypothesis for specificity and sensitivity for the study to be deemed a success.

The following null hypotheses will be tested independently for Day 1 and Day 2:

1. Sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth:

H0se: $\pi se = 0.20$ versus H1se: $\pi se \neq 0.20$

i.e., the lower bound of the two-sided 97.5% CI for sensitivity must be >0.20 (20% sensitivity) to conclude statistically significant success.

2. Specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth:

H0sp: $\pi sp = 0.85$ versus H1sp: $\pi sp \neq 0.85$

i.e., the lower bound of the two-sided 97.5% CI for specificity must be >0.85 (85% specificity) to conclude statistically significant success.

Where πse and πsp are the true sensitivity and specificity, respectively, for ^{64}Cu-SAR-bisPSMA PET.

Assuming a true sensitivity of 35% and a disease prevalence (i.e., positive detection by the Standard of Truth) of 25% a total of 324 participants will provide at least 80% power to reject the null hypothesis with a two-sided significance level of 2.5% (adjusted for multiplicity arising from independent assessments of Day 1 and Day 2 PET data). This sample size will provide > 94% power assuming the true specificity is 93%. This sample size is based on methods for exact Clopper-Pearson CIs.

To account for a drop-out and non-evaluable rate of 15% the study should aim to enroll a minimum of 383 participants.

11.4 Analysis Methods

11.4.1 Enrollment and Disposition

Screening data including reason(s) for screen failure will be listed for the All Participant Set. Enrollment and study completion data will be listed for the Enrolled Participant Set. The number of participants screened, number of participants enrolled, number of participants who completed the study and number of participants who withdrew from the study, along with the reason for discontinuation will be summarized using the All Participants Set and Enrolled Participant Sets.

The assignment to the Evaluable Set, Safety Set, Blood/Plasma PK Set and Urine PK Set will be listed and summarized for the Enrolled Participant Set. The reason for exclusion from an analysis set will be listed.

11.4.2 Demographic and Other Baseline Characteristics

Demographic and baseline characteristics will be listed and summarized descriptively for the Full Analysis Set and the Safety Set. The following demographic and baseline data will be summarized: age, race, ethnicity, height, weight and body mass index (BMI). Age is calculated from date of birth to date of informed consent. BMI is calculated from height and weight recorded at baseline.

11.4.3 Prior and Concomitant Medication

Prior and concomitant medications will be recorded, and Anatomical/Therapeutic/Chemical (ATC) class and preferred name coded using the World Health Organization drug dictionary (WHODrug). Prior medications are those medications started before the first administration of ^{64}Cu-SAR-bisPSMA. Concomitant medications are those medications taken after the administration of ^{64}Cu-SAR-bisPSMA. Prior medications that continued after the use of study drug are also considered concomitant medications. Prior and concomitant medications will be summarized for the Safety Set by ATC class (level 3) and preferred name.

11.4.4 Medical History

Medical history collected at screening will be coded using Medical Dictionary for Regulatory Activities (MedDRA). Medical history will be summarized for the Safety Set by system organ class and preferred term.

11.4.5 Germline Information

Germline information will be listed for all participants with this information available in the Full Analysis Set and summarized.

11.4.6 Prostate Cancer History

Prior PC history data, including date of diagnosis, tumor type, Gleason Score, ISUP Grade Group, clinical TNM staging, risk category, number of positive cores and locations of positive cores, PSA level, and testosterone level will be listed and summarized for the Full Analysis Set.

11.4.7 Institutional Standard of Care Imaging

Institutional SOC imaging will be performed during screening. The type of scan, and the results for each region and subregion will be listed and summarized for the Full Analysis Set (or amongst participants with evaluable data in FAS as applicable).

11.4.8 Study Drug Administration

All ^{64}Cu-SAR-bisPSMA administrations will be listed with complete information from the ^{64}Cu-SAR-bisPSMA administration eCRF.

11.4.9 ^{64}Cu-SAR-bisPSMA PET/CT Imaging

All ^{64}Cu-SAR-bisPSMA PET/CT imaging information for both Day 1 and Day 2 will be listed. For each timepoint, the number of lesions in each region/subregion will be summarized for the Full Analysis Set for each of the central readers.

The time from ^{64}Cu-SAR-bisPSMA administration to acquisition of the Day 1 and Day 2 PET/CT will be summarized for the Full Analysis Set.

11.4.10 Protocol Deviations

All participant level protocol deviations will be listed. Participant level protocol deviations will also be tabulated by type of deviation and Sponsor defined major deviations will be summarized separately by type of deviation for the Enrolled Participants Set.

11.4.11 Safety Analysis

All safety analyses will be conducted on the Safety Set. Where applicable, safety evaluations will be summarized by visit.

11.4.11.1 Adverse Events

AEs will be coded using the latest version of MedDRA. A TEAE is any AE that occurs after the administration of ^{64}Cu-SAR-bisPSMA. TEAEs, TEAEs related to investigational product, TEAEs with grade 3 or higher, TEAEs by severity, and treatment-emergent SAEs will be tabulated by MedDRA system organ class and preferred term.

All AEs, SAEs, AEs leading to study discontinuation and AEs leading to death will be listed by participant.

11.4.11.2 Clinical Laboratory Assessments

Hematology, biochemistry, urinalysis, eGFR, total PSA and testosterone measured values and change from baseline will be listed and summarized by visit.

11.4.11.3 Vital Signs

Vital signs measured values and change from baseline will be listed and summarized using descriptive statistics at all scheduled time points.

11.4.11.4 12-Lead ECG

12-Lead ECG observed values and change from baseline will be listed and summarized using descriptive statistics at all scheduled time points. The mean of the duplicate readings will be used for the summary and calculation of change from baseline.

Findings of 12-lead ECGs will be classified as either “normal”, “abnormal, not clinically significant”, or “abnormal, clinically significant” and categorial variables summarized with a shift table. For duplicate ECGs, the assessment of greatest clinical severity will be used.

The frequency of subjects with QTcF and QTcB pre dose and post dose values will be summarized according to the following categories: ≤450 msec, >450 and ≤ 480 msec, >480 and ≤ 500 msec, and >500 msec. In addition, the number and percentage of participants with the following maximum post-baseline increases in QTcF and QTcB will be summarized: <0 msec, 0 - 30 msec, >30 - 60 msec, > 60 msec.

11.4.11.5 Subgroup Safety Analysis

In addition to the planned safety analyses listed, where appropriate and where data allows, additional exploratory subgroup analyses will be undertaken to investigate renal and hepatic function on safety outcomes.

11.4.12 Efficacy Analysis

Primary efficacy analyses will be conducted with the Full Analysis Set, unless otherwise noted. Sensitivity/supportive analyses of the primary efficacy endpoints will be conducted with the Full Analysis Set and the Evaluable Set.

Determination of the Standard of Truths are described in Section 8.6.

The primary efficacy null and alternative hypotheses are presented in Section 11.3.

Within each imaging day, the type I error rate will be preserved by requiring that the null hypotheses for both sensitivity and specificity be rejected in order to draw the conclusion that ^{64}Cu-SAR-bisPSMA is efficacious for imaging. In addition, type I error will be further controlled by having the three readers from the central imaging laboratory be independent and the conclusion of efficacy will require that the same two of three readers meet the success criteria for sensitivity and specificity.

To account for independent assessments of the primary efficacy endpoints on Day 1 and Day 2 and control for multiplicity, the two-sided significance level (α) has been adjusted to 2.5% for assessment on each of the two imaging days.

11.4.12.1 Primary Efficacy Endpoints

The co-primary endpoints, on each of the ^{64}Cu-SAR-bisPSMA imaging days, are sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET/CT to detect PC within the pelvic LNs compared to the Standard of Truth.

The diagnostic performance of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LN region will be evaluated against the Standard of Truth. The result of the ^{64}Cu-SAR-bisPSMA PET and Standard of Truth will be matched for the left and right pelvic LN subregions. Diagnostic performance will be calculated at a patient level, that is, at least one PET-positive pelvic LN on the PET scan and one PC positive LN by histopathology on the same side of the pelvis (Left or Right) will be counted a True Positive at the patient level.

Sensitivity and specificity for pelvic LNs will be calculated per reader and per PET timepoint as follows:

Outcome (matched independently for left and right pelvic LN subregion)	**Pelvic LN SOT +**	**Pelvic LN SOT -**

Pelvic LN Region ^{64}Cu PET +	TP	FP
Pelvic LN Region ^{64}Cu PET -	FN	TN

Abbreviations: +=positive; -=negative; FN=false negative; FP=false positive; LN=lymph node; PET=positron emission tomography; SOT=Standard of Truth; TN= true negative; TP=true positive

Sensitivity: TP/(TP+FN) x 100

Specificity: TN/(TN+FP) x 100

For participants with missing pelvic LN Standard of Truth, the Standard of Truth will be imputed.

For participants who undergo ^{64}Cu-SAR-bisPSMA on only one of the two imaging days, the result for the missing imaging day will also be imputed. Full details of the imputation methods for the primary endpoints will be described in the SAP.

Assessment of the primary endpoints will be undertaken by determining the sensitivity and specificity point estimates and associated 2-sided 97.5% and 95.0% exact binomial CI [using the Clopper-Pearson method] for each of the three central readers and for Day 1 and Day 2 imaging, independently.

The null hypothesis for specificity (for Day 1 and Day 2, independently) will be rejected if the lower limit of the 97.5% CI exceeds 0.85 (i.e., 85%) for two of the three central readers. The null hypothesis for sensitivity (for Day 1 and Day 2, independently) will be rejected if the lower limit of the 97.5% CI exceeds 0.20 (i.e., 20%) for two of the three central readers. If specificity is a success, then the same two central readers need to reject the null hypothesis for sensitivity leading to overall success of the primary endpoint.

All ^{64}Cu-SAR-bisPSMA PET/CT for each central reader and the Standard of Truth for each participant will be listed.

All histopathology findings will be listed and summarized including number of lesions biopsied and results for the respective anatomical regions for the Evaluable Set.

11.4.12.2 Sensitivity Analysis for the Primary Efficacy Endpoints

Sensitivity analyses will be undertaken to investigate alternative imputation methods used for participants with missing Standard of Truth.

Full details of these methods will be described in the SAP.

11.4.12.3 Secondary Efficacy Endpoints

The secondary endpoints, on each of the ^{64}Cu-SAR-bisPSMA imaging days, are:

- positive predictive value (PPV) and negative predictive value (NPV) of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth (Determined for Full Analysis Set and Evaluable Set);
- sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate, pelvic LNs, extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth;
- sensitivity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the prostate gland compared to the Standard of Truth;

- sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LNs compared to the Standard of Truth, without the requirement of subregion (Left or Right) matching between ^{64}Cu-SAR-bisPSMA PET and Standard of Truth;
- inter-reader reliability of agreement estimated with a multiple-reader kappa statistic.

PPV for pelvic LNs will be calculated per reader and per PET timepoint as follows:

Outcome (matched independently for left and right pelvic LN subregion)	**Pelvic LN SOT+**	**Pelvic LN SOT -**
Pelvic LN Region ^{64}Cu PET +	TP	FP

Abbreviations: +=positive; -=negative; FP=false positive; LN=lymph node; PET=positron emission tomography; SOT=Standard of Truth; TP=true positive

PPV: TP/(TP+FP) x 100

NPV for pelvic LNs will be calculated per reader and per PET timepoint as follows:

Outcome (matched independently for left and right pelvic LN subregion)	**Pelvic LN SOT +**	**Pelvic LN SOT -**
Pelvic LN Region ^{64}Cu PET -	FN	TN

Abbreviations: +=positive; -=negative; FN=false negative; LN=lymph node; PET=positron emission tomography; SOT=Standard of Truth; TN= true negative

NPV: TN/(TN+FN) x 100

Sensitivity to detect PC within the prostate, pelvic LNs, extra-pelvic LNs, bone and visceral/soft tissue regions will be calculated per reader and per PET timepoint as follows:

Outcome	**Any region SOT +**
^{64}Cu PET + based on any matched region	TP
All regions ^{64}Cu PET - or ^{64}Cu PET- on any of matched region	FN

Abbreviations: +=positive; -=negative; FN=false negative; PET=positron emission tomography; SOT=Standard of Truth; TP=true positive

Sensitivity: TP/(TP+FN) x 100
Sensitivity for the prostate gland will be calculated per reader and per PET timepoint as follows:

Outcome	**Prostate SOT +**
Prostate Region ^{64}Cu PET +	TP
Prostate Region ^{64}Cu PET -	FN

Abbreviations: +=positive; -=negative; FN=false negative; PET=positron emission tomography; SOT=Standard of Truth; TP=true positive

Sensitivity: TP/(TP+FN) x 100

Sensitivity and specificity of ^{64}Cu-SAR-bisPSMA PET to detect PC within the pelvic LN region compared to the Standard of Truth, without the requirement of subregion (Left or Right) matching between ^{64}Cu-SAR-bisPSMA PET and Standard of Truth will be calculated per reader and per PET timepoint. In contrast to the primary efficacy analysis, participants will be considered True Positive if there is at least one PET-positive pelvic LN on the PET scan and one PC positive LN by histopathology, regardless of whether the subregions for the positive lesion(s) match. All other methodology applied to the primary efficacy endpoint analysis will be implemented (excluding the supportive/sensitivity analyses).

Assessment of the secondary endpoints will be undertaken by determining and tabulating the PPV, NPV and sensitivity point estimates and associated 2-sided 95.0% exact Clopper-Pearson binomial CI for each of the three central readers and for Day 1 and Day 2 imaging, independently.

To assess the consistency of ^{64}Cu-SAR-bisPSMA PET/CT interpretations for the three central readers for the primary endpoint, inter-reader reliability of agreement will be estimated with a multiple-reader kappa statistic and associated 95% CI.

11.4.12.4 Exploratory Efficacy Endpoints

On each of the ^{64}Cu-SAR-bisPSMA imaging days the following will be determined and tabulated:

- the number and percentage of participants who are SOC imaging pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive;
- the number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC imaging pelvic LN negative and histopathology negative;
- the number and percentage of participants who are SOC PSMA PET pelvic LN negative, but ^{64}Cu-SAR-bisPSMA PET pelvic LN positive and histopathology positive;
- the number and percentage of participants who are ^{64}Cu-SAR-bisPSMA PET pelvic LN positive but SOC PSMA PET pelvic LN negative and histopathology negative;
- the percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC imaging;
- the percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC imaging.
- the percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET compared to the SOC PSMA PET.
- the percentage of participants upstaged to N1 or M1 based on the ^{64}Cu-SAR-bisPSMA PET and confirmed as true positive by histopathology compared to the SOC PSMA PET.

On each of the ^{64}Cu-SAR-bisPSMA imaging days, PPV of ^{64}Cu-SAR-bisPSMA PET to detect distant metastatic PC within the extra-pelvic LNs, bone and visceral/soft tissue regions compared to the Standard of Truth will be calculated, as follows:

Outcome	**Extra-pelvic LN / Visceral/soft tissue / Bone SOT +**	**Extra-pelvic LN / Visceral/soft tissue / Bone SOT-**
Extra-pelvic LN / Visceral/soft tissue / Bone Region ^{64}Cu PET +	TP	FP

Abbreviations: +=positive; -=negative; FP=false positive; LN=lymph node; PET=positron emission tomography; SOT=Standard of Truth; TP=true positive

PPV: TP/(TP+FP) x 100

Assessment will be undertaken by determining and tabulating the PPV point estimates and associated 2-sided 95.0% exact Clopper-Pearson binomial CIs for each of the three central readers and for Day 1 and Day 2 imaging, independently.

On each of the ^{64}Cu-SAR-bisPSMA imaging days, the number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to baseline SOC imaging will be determined and tabulated.

On each of the ^{64}Cu-SAR-bisPSMA imaging days, relationship between PET-positivity (SUV, TBR), versus true/false positivity and as a function of region and lesion size will be determined and summarized. Relationship between PET-positivity (SUV, TBR) and Gleason Grade from the prostatectomy and pelvic LN dissection specimens will also be determined and summarized.

On each of the ^{64}Cu-SAR-bisPSMA imaging days, the lesion SUV, TBR and size on ^{64}Cu-SAR-bisPSMA PET compared to SOC PSMA PET will be determined and tabulated.

SUV, size and TBR in lesions will be listed and summarized by imaging day for the ^{64}Cu-SAR-bisPSMA PET and for the SOC PSMA PET.

The total number and difference in the total number of lesions detected per participant on Day 1 and Day 2 ^{64}Cu-SAR-bisPSMA PET/CT scans will be summarized descriptively by central reader. The number of lesions at Day 1 and Day 2 will be compared and p-value and mean difference summarized by central reader.

On each of the ^{64}Cu-SAR-bisPSMA imaging days, sub-group analysis will be undertaken and summarized for each group based on group classification for PSA at baseline, testosterone at baseline, Gleason Grade pre-prostatectomy, stage, grade, risk and germline information for the following variables:

- sensitivity and specificity to detect PC within the pelvic LNs;
- PPV and NPV to detect PC within the pelvic LNs;
- sensitivity to detect PC within the participants overall;
- sensitivity to detect PC within the prostate gland;
- PPV to detect distant metastatic PC;
- number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC imaging;

- number of lesions detected per participant and per region (prostate gland, pelvic LNs, extra-pelvic LNs, bone, visceral/soft tissue) on ^{64}Cu-SAR-bisPSMA PET compared to SOC PSMA PET.

Exploratory subgroup analysis will be undertaken to investigate the effect of ^{64}Cu-SAR-bisPSMA PET acquisition time relative to dose administration on the primary efficacy endpoints and number of lesions detected.

In addition to the planned efficacy analyses described, where appropriate and where data allows, additional exploratory subgroup analyses will be undertaken to investigate renal and hepatic function on efficacy outcomes.

11.4.13 Pharmacokinetic Analysis

Individual blood ^{64}Cu-SAR-bisPSMA concentration data will be listed and summarized by protocol specified collection timepoints, with descriptive statistics including the CV% and geometric mean. Individual and mean concentration-time profiles will also be presented graphically on linear and log-linear scales.

Where data are sufficient for parameter determination, the following blood ^{64}Cu-SAR-bisPSMA PK parameters will be estimated using standard non-compartmental procedures:

- C_{max}: Maximum observed concentration;
- T_{max}: Time to reach the maximum concentration;
- $AUC_{0\text{-last}}$: Area under the plasma concentration-time curve from the time of the dose administration to the time of the last measurable concentration calculated by the linear up log down trapezoidal method;
- $AUC_{0\text{-inf}}$: Area under the concentration-time curve from the time of the dose administration to time infinity calculated by the linear up log down trapezoidal method;
- kel: Terminal elimination rate constant, estimated by linear regression of the logarithmically transformed concentration-time data;
- $t_{1/2e}$: Elimination half-life associated with the terminal slope (kel) of the semilogarithmic concentration-time curve, calculated as 0.693/kel;
- CL: The apparent total clearance;
- V_z: The apparent total volume of distribution.

Blood ^{64}Cu-SAR-bisPSMA PK parameters will be listed and summarized with descriptive statistics including, %CV, geometric mean and geometric %CV. Further details on terminal elimination phase regression methods and acceptance criteria will be presented in the SAP.

Ratios of blood-to-plasma radioactivity at each time point will be listed and summarized by timepoint.

Individual urine ^{64}Cu-SAR-bisPSMA amount excreted data, expressed as amount and percentage of total administered dose will be listed and summarized by protocol specified collection interval with descriptive statistics including the CV% and geometric mean. Individual and mean urine excreted amount-time profiles will also be presented graphically.

Where data are sufficient for parameter determination the following urine ^{64}Cu-SAR-bisPSMA PK parameters will be estimated using standard non-compartmental procedures:

- T_{Rmax}: Midpoint of the collection interval associated with the maximum observed excretion rate;
- R_{max}: Maximum observed excretion rate, at time T_{Rmax};
- Ae_{0-t}: Cumulative amount recovered in urine from time zero to time of last collection interval, calculated as the sum of the amounts excreted over each collection interval;
- $\%FE_{0-t}$: Fraction (percent) of total dose recovered in urine.

Urine ^{64}Cu-SAR-bisPSMA PK parameters will be listed and summarized with descriptive statistics including %CV, geometric mean and geometric %CV.

In addition to the planned PK analyses described, where appropriate and where data allows, additional exploratory subgroup analyses will be undertaken to investigate renal and hepatic function on ^{64}Cu-SAR-bisPSMA PK.

11.4.14 Dosimetry Analysis

The dosimetry analysis will be conducted using the Dosimetry Set. Dosimetry will be assessed using the ^{64}Cu-SAR-bisPSMA PET/CT scans. The following data will be summarized using descriptive statistics:

- Absorbed dose per unit administered activity (mGy/MBq) in organs from ^{64}Cu-SAR-bisPSMA;
- Effective dose per unit administered activity (mSv/MBq) from ^{64}Cu-SAR-bisPSMA;
- Effective dose (mSv) from 200 MBq of ^{64}Cu-SAR-bisPSMA.

11.5 Interim Analysis

There is no interim analysis planned for this study.

11.6 Handling of Missing Data

Missing disposition, safety (except for missing data about AE relationship to ^{64}Cu-SAR-bisPSMA, these will be imputed as ‘related to ^{64}Cu-SAR-bisPSMA’) and PK data will not be imputed. Imputation methods for the primary efficacy analyses, and sensitivity analyses for assessment of the imputation method are described in the SAP.

12 STUDY ADMINISTRATION AND RESPONSABILITIES

12.1 General Investigator Responsibilities

The Investigator must ensure that:

- He or she will personally conduct or supervise the study;
- His or her staff and all persons who assist in the conduct of the study clearly understand their responsibilities and have their names included in the protocol specific Study Personnel Responsibility/Signature Log;
- The study is conducted according to the protocol and all applicable regulations;
- The protection of each participant's rights and welfare is maintained;
- Signed and dated informed consent and, when applicable, permission to use protected health information are obtained from each participant before conducting study procedures. If a participant withdraws permission to use protected health information, the Investigator will obtain a written request from the participant and will ensure that no further data will be collected from the participant;
- The consent process is conducted in compliance with all applicable regulations and privacy acts;
- The IRB complies with applicable regulations and conducts initial and ongoing reviews and approvals of the study;
- Any amendment to the protocol is submitted promptly to the IRB;
- Any significant protocol deviations are reported to the Medical Monitor, the Sponsor, and the IRB according to the guidelines at each study site;
- Electronic case report form pages are completed within 5 days of each participant's visit (unless required earlier for SAE reporting);
- All SAEs and pregnancies are reported to the Sponsor or designee within 24 hours of knowledge and to the IRB per IRB requirements;
- All safety reports are submitted promptly to the IRB, as per IRB requirements.

12.2 Protocol Compliance

The Investigator is responsible for ensuring the study is conducted in accordance with the procedures and evaluations described in this protocol.

12.3 Compliance with Ethical and Regulatory Guidelines

The Investigator will ensure that this study is conducted in accordance with International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use, or with the laws and regulations of the country in which the research is conducted, whichever affords the greater protection to the study participant. For studies conducted under a US Investigational New Drug (IND) application, the Investigator will ensure adherence to the basic principles of GCP as outlined in 21 Code of Federal Regulations (CFR) 312, subpart D, "Responsibilities of Sponsors and Investigators"; 21 CFR, Part 50, 1998; and 21 CFR, Part 56, 1998.

This study is also subject to and will be conducted in accordance with 21 CFR, Part 320, 1993, "Retention of Bioavailability and Bioequivalence Testing Samples."

Because this is a "covered" clinical study, the Investigator will ensure adherence to 21 CFR, Part 54, 1998; a covered clinical study is any "study of a drug or device in humans submitted in a marketing application or reclassification petition subject to this part that the applicant or FDA relies on to establish that the product is effective (including studies that show equivalence to an effective product) or that make a significant contribution to the demonstration of safety." This requires that Investigators and all sub-Investigators must provide documentation of their financial interest or arrangements with the Sponsor, or proprietary interests in the drug being studied. This documentation must be provided before participation of the Investigator and any sub Investigator in the study. The Investigator or sub Investigator agrees to notify the Sponsor of any change in reportable financial interests during the study and for 1 year following completion of the study. Study completion is defined as the date that the last participant has completed the protocol-defined activities.

12.4 Institutional Review Board

This protocol and any accompanying material to be provided to the participant (such as advertisements, participant information sheets, or descriptions of the study used to obtain informed consent) will be submitted by the Investigator to an IRB. Approval from the IRB must be obtained before starting the study and should be documented in a letter from the IRB to the Investigator specifying the protocol number, protocol version, protocol date, documents reviewed, and date on which the committee met and granted the approval. A signed protocol approval page, a letter confirming IRB approval of the protocol and informed consent, and a statement that the IRB is organized and operates according to GCP and the applicable laws and regulations must be forwarded to the Sponsor before screening participants for the study. Additionally, US study sites must forward a signed Form FDA 1572 (Investigator Obligation Form) to the Sponsor before screening participants for study enrollment.

Any modifications or amendments made to the protocol or ICF after receipt of the initial IRB approval must also be submitted to the IRB for approval before implementation. Only changes necessary to eliminate apparent immediate hazards to the participants may be initiated prior to IRB approval. In that event, the Investigator must notify the IRB, the Medical Monitor, and the Sponsor in writing within 5 working days after implementation. If a change to the protocol in any way increases the risk to the participant or changes the scope of the study, then written documentation of IRB approval must be received by the Sponsor before the amendment may take effect. Additionally, under this circumstance, information on the increased risk and/or change in scope must be provided to participants already actively participating in the study, and they must read, understand, and sign any revised informed consent document confirming willingness to remain in the study.

The Investigator must ensure that all local regulations and reporting requirements to the IRB are met.

12.5 Informed Consent Process

Note: All references to "participant" in this section refer to the study participant or his legally authorized representative.

The Sponsor (or its designee) will provide Investigators with a multicenter ICF for this study. Investigators may adapt the information to suit the needs of their institution, if necessary (although it must reflect the required elements of informed consent specified in 21 CFR Part

50.25). The final ICFs must be accepted by the Sponsor and approved by the IRB. Investigators must provide the Sponsor with an unsigned copy of the final ICFs before and after it is approved by the IRB. If any new information becomes available that might affect participants' willingness to participate in the study, or if any amendments to the protocol require changes to the ICF, the Sponsor will provide Investigators with a revised ICF.

Prior to participating in any study-related procedure, each participant must sign and date an IRB-approved ICF written in a language the participant can understand. The ICF should be as nontechnical as practical and understandable to the participant. The ICF must provide the participant with all the information necessary to make an informed decision about their participation in the study, including the nature and intended purpose of the study, possible benefits, possible risks, and disclosures of the participant's personal information and personal health information for purposes of conducting the study. The ICF will include details of the requirements of the participant and the fact that he/she is free to withdraw at any time without giving a reason and without prejudice to their further medical care. Before informed consent is obtained, the participant should be given ample time and opportunity to inquire about the details of the study. All questions must be answered to the satisfaction of the participant.

Once signed, the original ICF will be stored in the Investigator's site file and made available for review by the Sponsor. Documentation of the informed consent discussion must be noted in the participant's case history. All participants will receive a copy of their signed and dated ICF.

If the ICF is revised during the study and requires the participant to be re consented, informed consent will be obtained in the same manner as for the original ICF.

12.6 Confidentiality

Every effort will be made to maintain the anonymity and confidentiality of all participants during this clinical study. However, because of the experimental nature of this investigational product, the Investigator agrees to allow the IRB, representatives of the Sponsor and its designated agents, and authorized employees of appropriate regulatory agencies to inspect the facilities used in this study and, for purposes of verification, allow direct access to the study site records of all participants enrolled into this study. This includes providing by fax, e-mail, or regular mail de-identified copies of clinical, laboratory, ECG, radiology, pathology, and/or other test results when requested by the Sponsor. A statement to this effect will be included in the ICF and a permission form authorizing the use of protected health information will also be included.

In accordance with local and national participant privacy regulations, the Investigator or designee must explain to each participant that in order to evaluate study results, the participant's protected health information obtained during the study may be shared with IRBs, the Sponsor and its designees, and regulatory agencies. It is the Investigator's or designee's responsibility to obtain written permission to use protected health information from each participant. If a participant withdraws permission to use protected health information, it is the Investigator's responsibility to obtain the withdrawal request in writing from the participant and to ensure that no further data will be collected from the participant. Any data collected on the participant before withdrawal will be used in the analysis of study results. The Sponsor will only use or disclose the participant's protected health information consistent as defined in the ICF.

The Investigator must assure that each participant's anonymity will be strictly maintained, and that each participant's identity is protected from unauthorized parties. Only participant initials,

date of birth, and an identification code (but no participant names) should be recorded on any form or biological sample submitted to the IRB, to the Sponsor or its designees (e.g., laboratories), or to regulatory authorities. However, sufficient information must be retained at the study site to permit sample data and data in the database to be connected with the unique participant number assigned to each study participant.

The Investigator agrees that all information received from the Sponsor, including, but not limited to, the investigational product, the IB, this protocol, the eCRFs, and any other study information remain the sole and exclusive property of the Sponsor during the conduct of the study and thereafter. This information is not to be disclosed to any third party (except employees or agents directly involved in the conduct of the study or as required by law) without prior written consent from the Sponsor. The Investigator further agrees to take all reasonable precautions to prevent the disclosure by any employee or agent of the study site to any third party or otherwise into the public domain.

12.7 Study Files and Retention of Records and Biological Samples

The Investigator must maintain adequate and accurate records to enable the conduct of the study to be fully documented and the study data to be subsequently verified by the IRB, representatives of the Sponsor and its designated agents, and authorized employees of appropriate regulatory agencies. These documents should be classified into at least the following 2 categories: Investigator's study file and participant clinical source documents.

The Investigator's study file will contain the protocol/amendments, eCRF and query forms, the IRB and governmental approval with correspondence, signed ICF, drug accountability records, staff curriculum vitae and authorization forms (e.g., Form FDA 1572), and other appropriate documents and correspondence pertaining to the conduct of the study.

The required source data referenced in the monitoring plan for the study should include sequential notes containing at least the following information for each participant:

- Participant identification (name, date of birth, sex, race and ethnicity);
- Documentation that participant meets each of the eligibility criteria, e.g., history and confirmation of diagnosis (to support inclusion and exclusion criteria);
- Participation in study (including study number);
- Study discussed and date of informed consent/re-consent;
- Dates of all visits;
- Documentation that protocol-specific procedures were performed;
- Results of efficacy parameters, as required by the protocol;
- Start and end date (including dose regimen) of investigational products (including relevant drug dispensing information);
- Record of all AEs and other safety parameters (including start and end date, causality, and severity [intensity]);
- Concomitant medications (including start and end date and dose if relevant dose changes occur);
- Date of study completion and reason for discontinuation, if applicable.

All clinical study documents must be retained by the Investigator until at least 2 years after the last approval of a marketing application in an ICH region (the United States, Europe, or Japan) and until there are no pending or contemplated marketing applications in an ICH region; or, if no application is filed or if the application is not approved for such indication, until 2 years after the investigation is discontinued and regulatory authorities have been notified or for 15 years, whichever is longer. Investigators may be required to retain documents longer if required by applicable regulatory requirements, by local regulations, or by an agreement with the Sponsor. The Investigator must notify the Sponsor and obtain written approval from the Sponsor before destroying any clinical study records. The Investigator will promptly notify the Sponsor in the event of accidental loss or destruction of any study records. The Sponsor will inform the Investigator of the date that study records may be destroyed or returned to the Sponsor.

The Sponsor must be notified in advance and must provide express written approval of any change in the maintenance of the foregoing documents if the Investigator wishes to move study records to another location or assign responsibility for record retention to another party.

If the Investigator cannot guarantee this archiving requirement at the study site for any or all of the documents, special arrangements must be made between the Investigator and the Sponsor to store these in sealed containers outside of the study site so that they can be returned sealed to the Investigator in case of a regulatory audit. When source documents are required for the continued care of the participant, appropriate copies should be made for storage outside of the study site.

Biological samples should be stored according to institutional guidelines, aligned with state and federal recommendations, e.g. a minimum of 2 years for specimen blocks and 10 years for histology slides.. The Investigator should promptly notify the Sponsor in the event of accidental loss or destruction of any study samples. With the permission of the Sponsor, the retained samples may be transferred to an acceptable designee, such as another Investigator, another institution, a contract storage site, or to the Sponsor.

12.8 Participant Screening Log

The Investigator must keep a record that lists all participants who signed the ICF (including those who did not undergo screening assessments). For those participants who declined to participate, post signing the ICF or were subsequently excluded from enrollment, the reasons for not enrolling in the study must be described.

12.9 Modifications of the Protocol or Informed Consent Documents

Protocol modifications, except those intended to reduce immediate risk to study participants, will be made only by the Sponsor. All protocol modifications must be submitted to the IRB in accordance with local requirements. Except as noted in Section 12.4, IRB approval must be obtained before changes can be implemented.

The ICF cannot be changed without prior written approval by the Sponsor and the study site's IRB.

The Sponsor may not reimburse the Investigator for cases in which study procedures and evaluations are not conducted in line with the IRB approved protocol.

12.10 Case Report Forms

Authorized study site personnel will complete eCRFs designed for this study according to the completion guidelines that will be provided. An eCRF is required and must be completed for each enrolled participant, with all required study data accurately recorded such that the information matches the data contained in medical records (e.g., physicians' notes, nurses' notes, study site charts, or other study-specific source documents). The Investigator will ensure that the eCRFs are accurate, complete, legible, and completed within 5 days of each participant's visit (unless required earlier for SAE reporting). The Investigator will ensure that source documents that are required to verify the validity and completeness of data transcribed on the eCRFs are never obliterated or destroyed. As required by the protocol, eCRFs should also be completed for those participants who fail to complete the study (even during the screening period). If a participant withdraws from the study, the reason must be noted on the eCRF and thorough efforts should be made to clearly document outcome.

The eCRFs for this study will exist within a web-based electronic data capture (EDC) system. After the Investigator or the Investigator's designees (e.g., research coordinators) have been appropriately trained, they will be given access to the EDC system and will enter the data required by the protocol into the EDC system. Any change of data will be made via the EDC system, with all changes tracked by the system to provide an audit trail.

The eCRF must be completed and signed by the principal Investigator or sub-Investigator (as appropriate) within a reasonable time period after data collection. This signature serves to attest that the information contained in the eCRF is true.

12.11 Clinical Monitoring

Representatives of the Sponsor or its designee will monitor this study until completion. Monitoring will be conducted through personal visits with the Investigator and study site staff as well as any appropriate communications by mail, e-mail, or telephone. The monitor is responsible for routine review of the eCRFs at regular intervals throughout the study to verify adherence to the protocol and the completeness, consistency, and accuracy of the data being entered on them. The purpose of monitoring is to ensure compliance with the protocol and the quality and integrity of the data.

In accordance with GCP, the study monitor must have direct access to the Investigator's source documentation in order to verify the data recorded in the eCRFs for consistency. The Investigator agrees to cooperate with the monitor to ensure that any problems detected in the course of these monitoring visits are resolved.

12.12 Inspections

The source documents for this study must be made available to appropriately qualified personnel from the Sponsor or its representatives, to the IRB, and to regulatory authority or health authority inspectors as a part of their responsibility to protect human participants in research. The Investigator agrees to provide access to records, facilities, and personnel for the effective conduct of any inspection or audit to representatives of the Sponsor and regulatory agencies. It is important that the Investigator and relevant institutional personnel are available during monitoring visits and possible audits or inspections and that sufficient time is devoted to the process. If the Investigator is notified of an inspection by a regulatory authority, the Investigator agrees to notify the Sponsor immediately.

12.13 Data Management

Electronic data capture will be used to enter study data eCRFs and to transfer the data into a study-specific electronic database. Instances where the eCRF captured data are considered source data will be outlined in separate study specific documentation. During the data collection process, automated quality assurance programs will be used to identify missing data, out-of-range data, and other data inconsistencies. Requests for data clarification or correction will be forwarded to the investigative study site for resolution.

Quality assurance and quality control systems will be implemented and maintained according to written standard operating procedures to ensure that the data are generated, recorded, and reported in compliance with the protocol, GCP, and applicable regulatory requirements. Data collection and storage systems will provide an audit trail, security mechanisms, and electronic signature capabilities that meet the requirements of FDA Title 21 of CFR Part 11 regarding electronic records and electronic signatures.

Data security will be controlled through appropriate and specific restriction of access to only data and systems required by individual users to accomplish their roles in the data management process. Individual login and password protections will be employed at study sites and at the Sponsor or its designee. The database will exist on physically secured servers. Data backups will be done regularly and will be stored in separate facilities. Printed documents relating to the study will be secured when not under review.

12.14 Clinical Study Insurance

The Sponsor will secure clinical study insurance. An insurance certificate will be made available to the participating study sites before study initiation.

12.15 Communications with Regulatory Authorities

The Sponsor, working either directly or through designees, will assume responsibility for regulatory interactions with relevant regulatory authorities. The Sponsor will maintain an IND application for the investigational product in support of the study in the United States and will maintain similar regulatory applications with other regulatory authorities as required for conduct of the study. In fulfilling this responsibility, the Sponsor (or a designee) will collect, assemble, and communicate all required regulatory documents (e.g., Form FDA 1572, Investigator financial disclosure forms, protocol and protocol amendments, IB, informed consent documents, annual reports) as required by regulation. The Sponsor (or a designee) will also assume responsibility for safety reporting to regulatory authorities as described in Section 10.5.

12.16 Publication and Information Disclosure Policy

All information provided by the Sponsor and all data and information generated by the site as part of the study (other than a participant's medical records) are the sole property of Sponsor.

For clinical interventional studies in patients, Sponsor will post study results on websites such as https://clinicaltrials.gov/ in accordance with FDA reporting rules.

The results of this study may be published or presented at scientific meetings. If this is foreseen, the investigator agrees to submit all manuscripts or abstracts to the sponsor before submission. This allows the sponsor to protect proprietary information and to provide comments. Patient or institution specific publications (this includes but is not limited to manuscripts, presentations,

abstracts, posters) are only allowed after the main study results are analyzed and presented, or upon approval by the Sponsor.

The sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the sponsor will generally support publication of multicenter studies only in their entirety and not as individual site data. In this case, a coordinating investigator will be designated by mutual agreement.

Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

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14 INVESTIGATOR AGREEMENT

Protocol Number: CLP04

Protocol Title: Positron Emission Tomography using 64Cu-SAR-bisPSMA in participants with high-risk prostate cancer prior to radical prostatectomy: A prospective, single-arm, multi-center, blinded-review, Phase 3 diagnostic performance study

For an Investigator in the United States:

I have carefully read and understand the foregoing protocol and agree that it contains all the necessary information for conducting this study safely. I will conduct this study in strict accordance with this protocol, ICH guidelines for GCP, the CFR, the Health Insurance Portability and Accountability Act, and local regulatory guidelines. I will attempt to complete the study within the time designated. I will ensure that the rights, safety, and welfare of participants under my care are protected. I will ensure control of the drugs under investigation in this study. I will provide copies of the protocol and all other study-related information supplied by the Sponsor to all personnel responsible to me who participate in the study. I will discuss this information with them to assure that they are adequately informed regarding the drug and conduct of the study. I agree to keep records on all participant information (case report forms, shipment and drug return forms, and all other information collected during the study) and drug disposition in accordance with FDA regulations. I will not enroll any participants into this protocol until IRB approval and Sponsor approval are obtained.

_____

Site or Institution Name

_____

Site number

_____

Investigator Name (Print)

_____ _____

Investigator Signature Date

For an Investigator in Australia:

The investigator will conduct this study as detailed herein, in compliance with current standards for the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, Good Clinical Practice and the applicable regulatory requirements and will make every reasonable effort to complete the study within the time designated. The investigator will assure that no deviation from, or changes to the protocol will take place without prior agreement from the sponsor and documented approval from the Human Research Ethics Committee, except where necessary to eliminate an immediate hazard(s) to the study participants. This document contains information that is privileged or confidential. As such, it may not be disclosed unless specific prior permission is granted in writing by the Sponsor or such disclosure is required by federal or other laws or regulations. Persons to whom any of this information is to be disclosed must first be informed that the information is confidential. These restrictions on disclosure will apply equally to all future information supplied, which is indicated as privileged or confidential. The investigator agrees that all staff members involved in the conduct of this study are informed about their obligations in meeting the above commitments.

_____

Site or Institution Name

_____

Site number

_____

Investigator Name (Print)

_____ _____

Investigator Signature Date